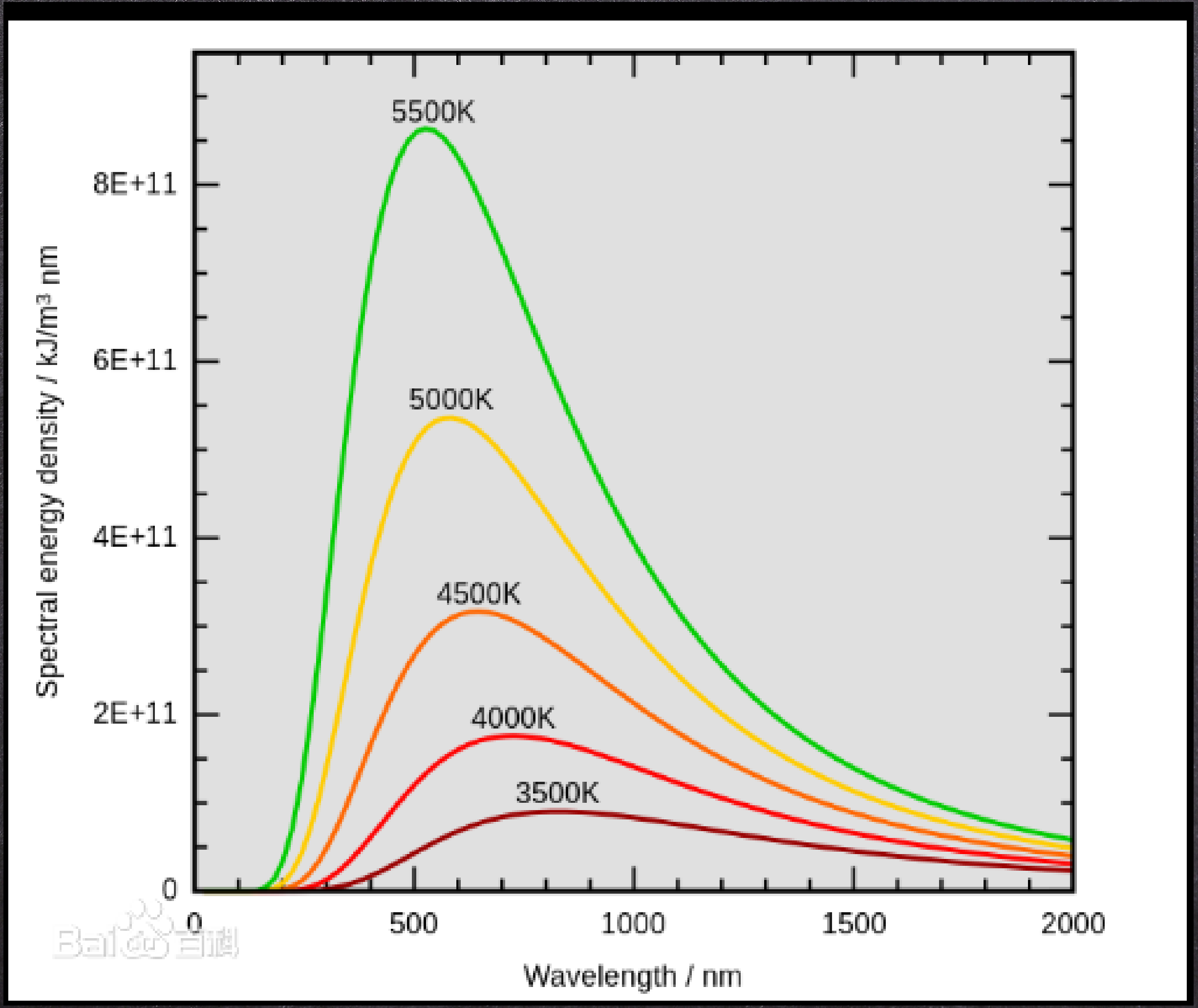
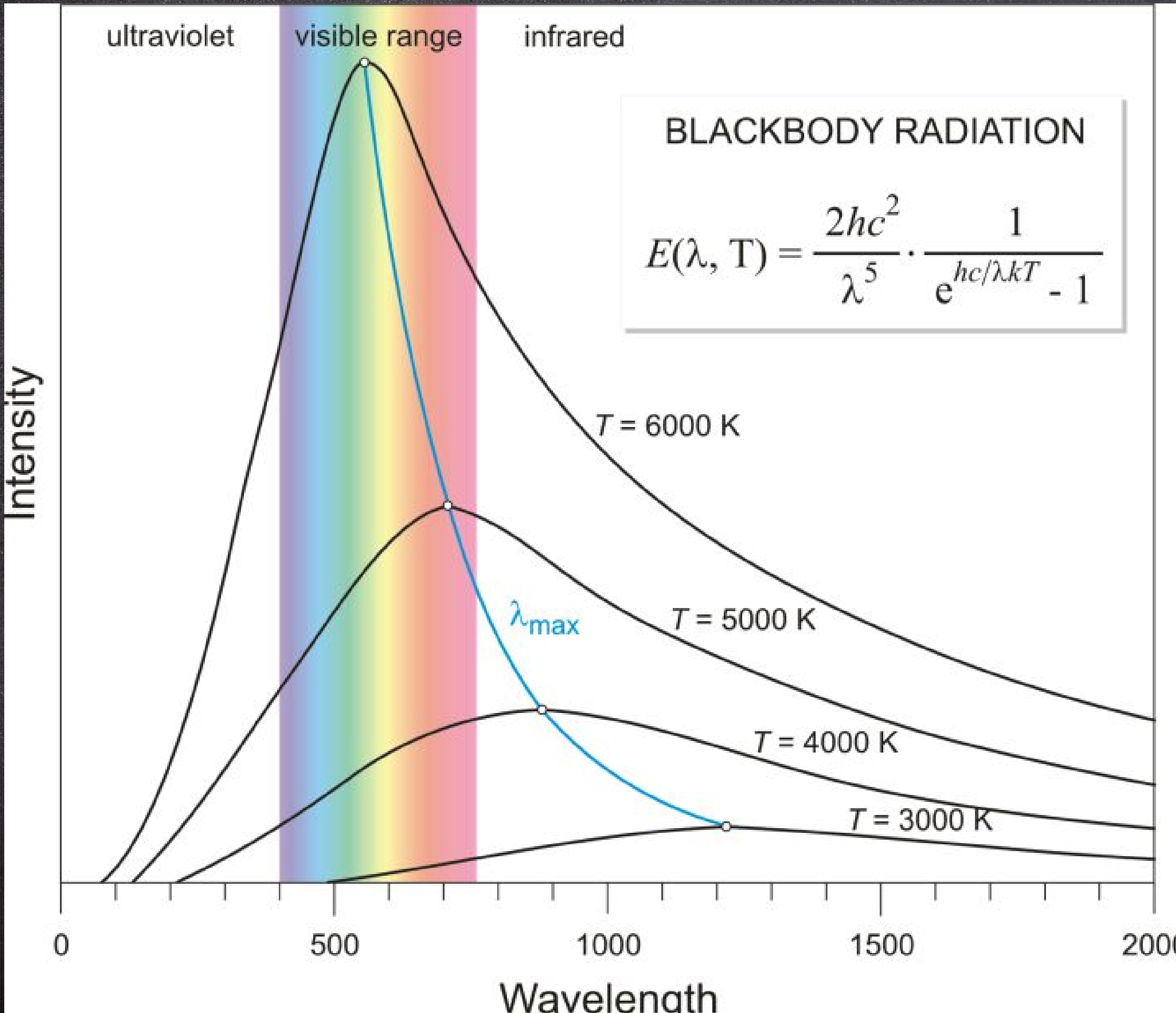
数字图像处理

黑体辐射


5500K
5000K
4500K
4000K
3500K
8E+11
6E+11
4E+11
2E+11
0
0
500
1000
1500
2000
Spectral energy density / kJ/m³ nm
Wavelength / nm



ultraviolet
visible range
infrared
BLACKBODY RADIATION
$$E(\lambda, T) = \frac{2hc^2}{\lambda^5} \cdot \frac{1}{e^{hc/\lambda kT} - 1}$$
T = 6000 K
T = 5000 K
T = 4000 K
T = 3000 K
λ_{max}
Intensity
0
500
1000
1500
2000
Wavelength

太阳与天空

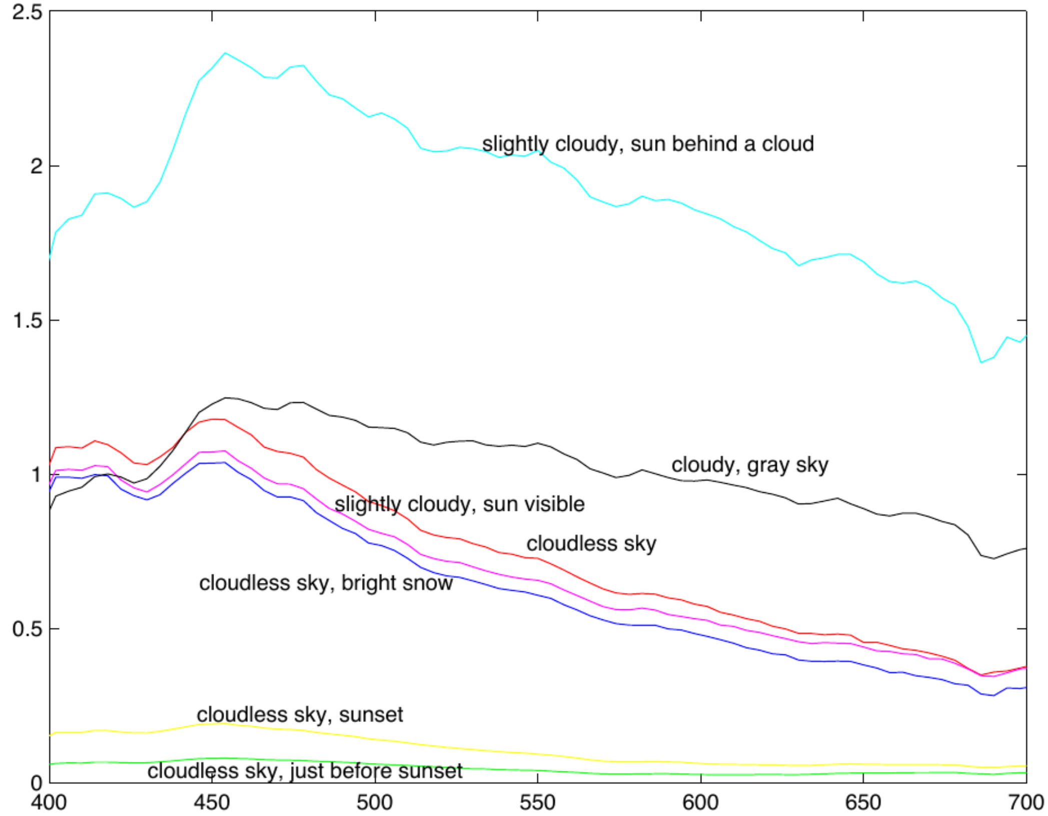


Figure 4.3. There are significant variations in the relative spectral power of daylight measured at different times of day and under different conditions. The figure shows a series of seven different daylight measurements, made by Jussi Parkkinen and Pertti Silfsten, of daylight illuminating a sample of barium sulphate (which gives a very high reflectance white surface). Plot from data obtainable at `http://www.it.lut.fi/research/color/lutcs_database.html`.

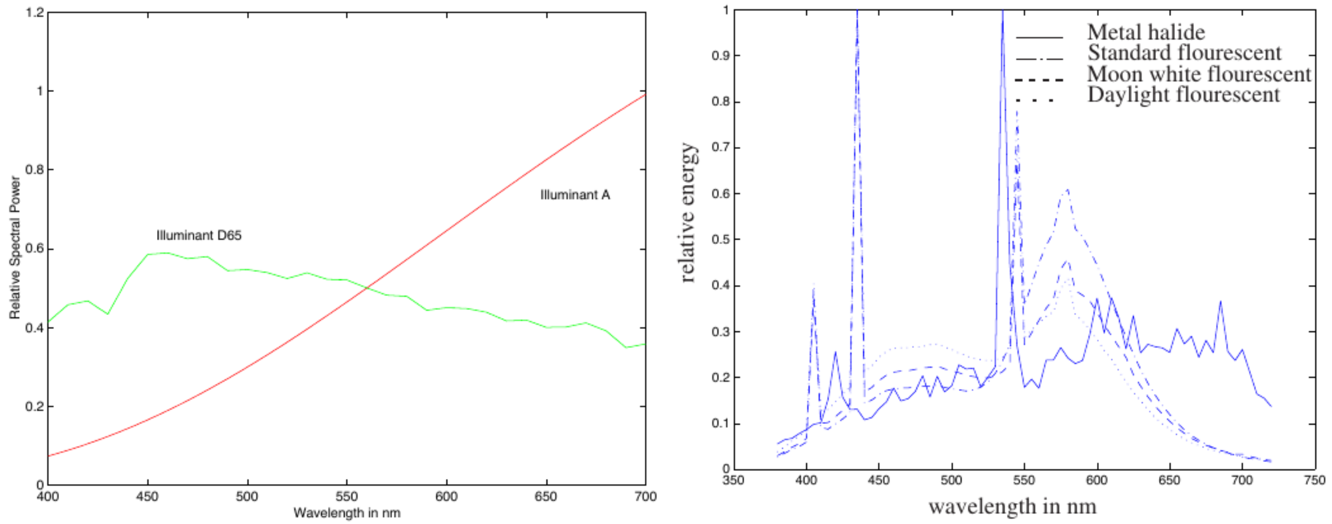


Figure 4.4. There is a variety of illuminant models; the graph shows the relative spectral power distribution of two standard CIE models, illuminant A — which models the light from a 100W Tungsten filament light bulb, with colour temperature 2800K — and illuminant D-65 — which models daylight. *Figure plotted from data available at* `http://www-cvrl.ucsd.edu/index.htm`.The relative spectral power distribution of four different lamps from the Mitsubishi Electric corp, measured by ****, data from *****. Note the bright, narrow bands that come from the flourescing phosphors in the fluorescent lamp.

光谱反射率——花与叶

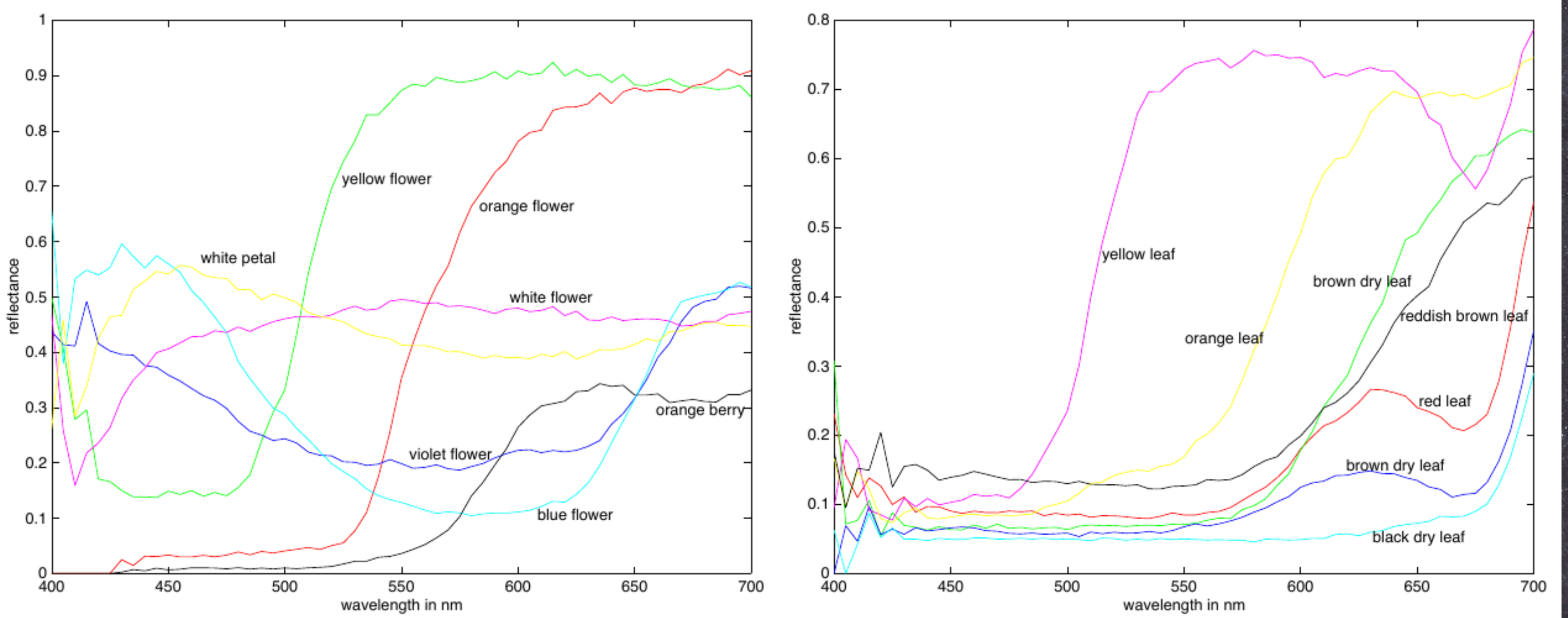


Figure 4.1. Spectral albedoes for a variety of natural surfaces, measured by Esa Koivisto, Department of Physics, University of Kuopio, Finland. On the left, albedoes for a series of different natural surfaces — a colour name is given for each. On the right, albedoes for different colours of leaf; again, a colour name is given for each. These figures were plotted from data available at `http://www.it.lut.fi/research/color/lutcs_database.html`.

光谱反射率——红花与绿叶

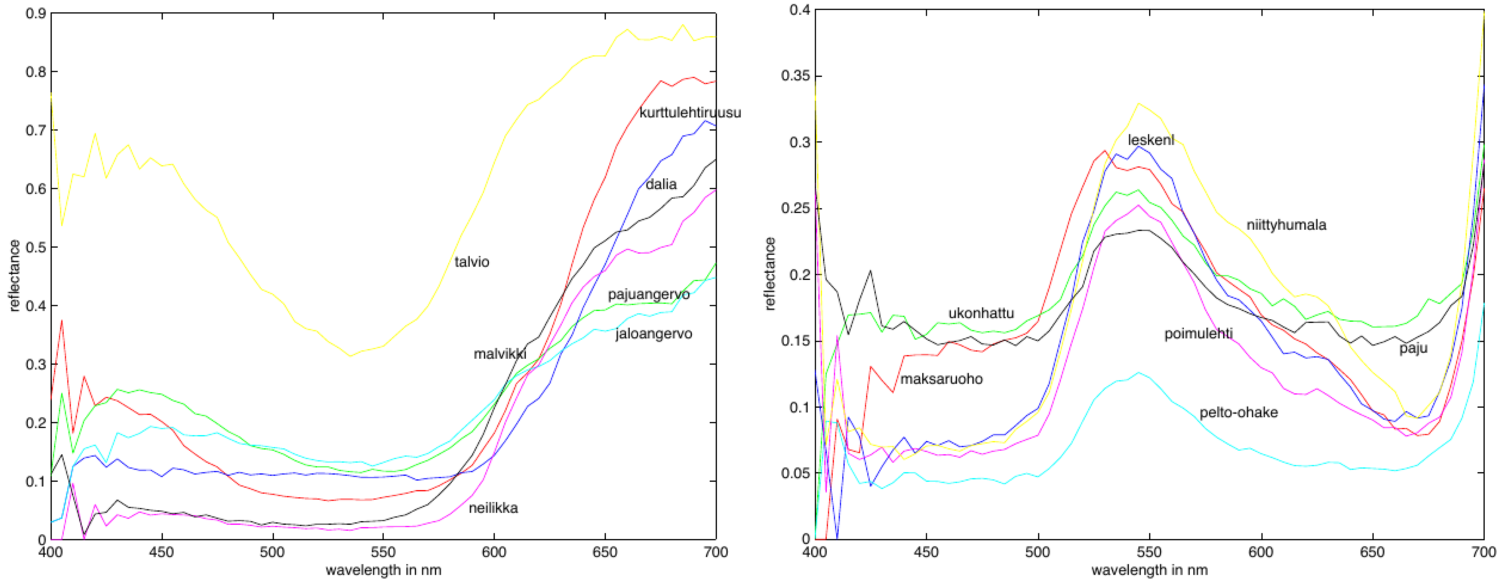
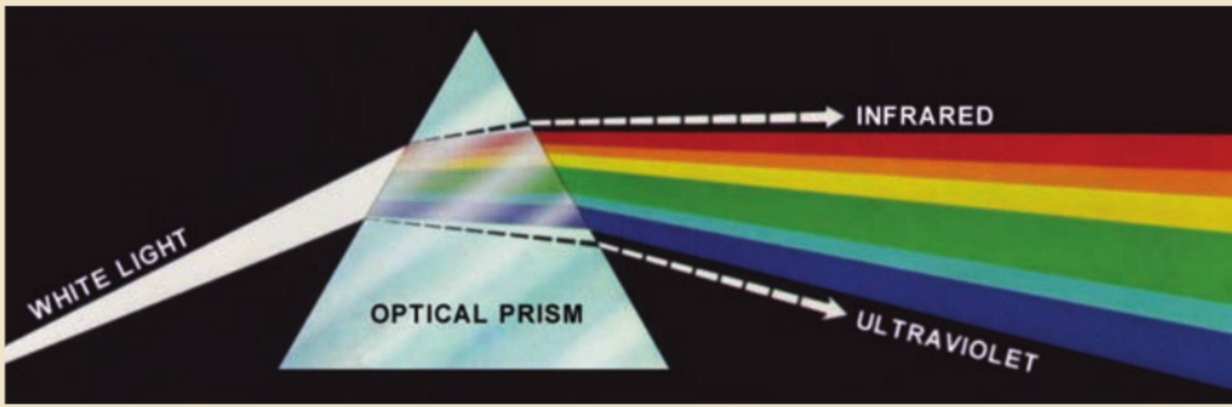


Figure 4.2. More spectral albedoes for a variety of natural surfaces, measured by Esa Koivisto, Department of Physics, University of Kuopio, Finland. On the left, albedoes for a series of different red flowers. Each is given its Finnish name. On the right, albedoes for green leaves; again, each is given its Finnish name. You should notice that these albedoes don't vary all that much. This is because there are relatively few mechanisms that give rise to colour in plants. These figures were plotted from data available at `http://www.it.lut.fi/research/color/lutcs_database.html`.

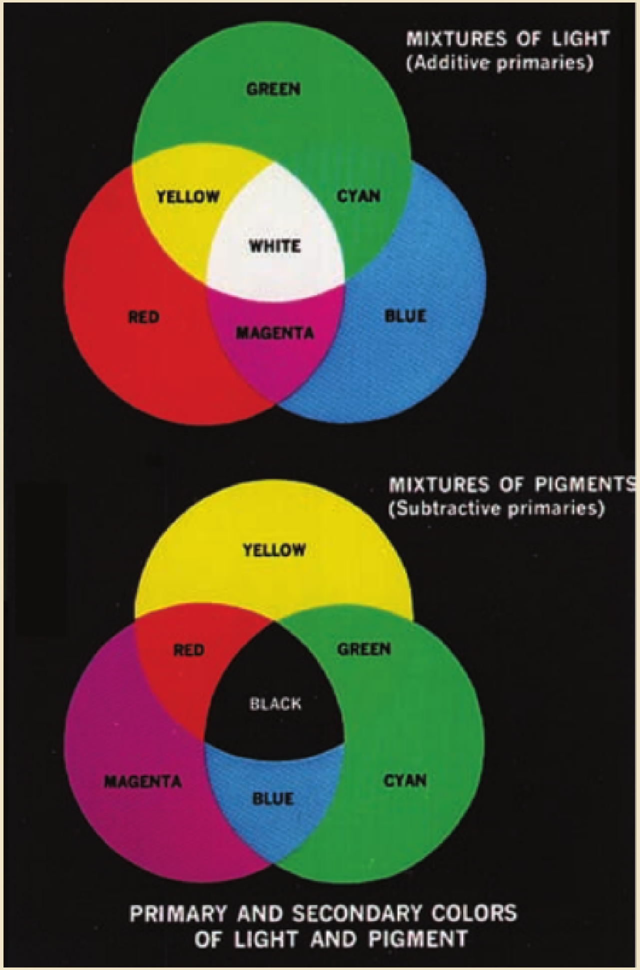
光谱

FIGURE 6.1
Color spectrum seen by passing white light through a prism. (Courtesy of the General Electric Co., Lighting Division.)



a
b

FIGURE 6.4 Primary and secondary colors of light and pigments. (Courtesy of the General Electric Co., Lighting Division.)



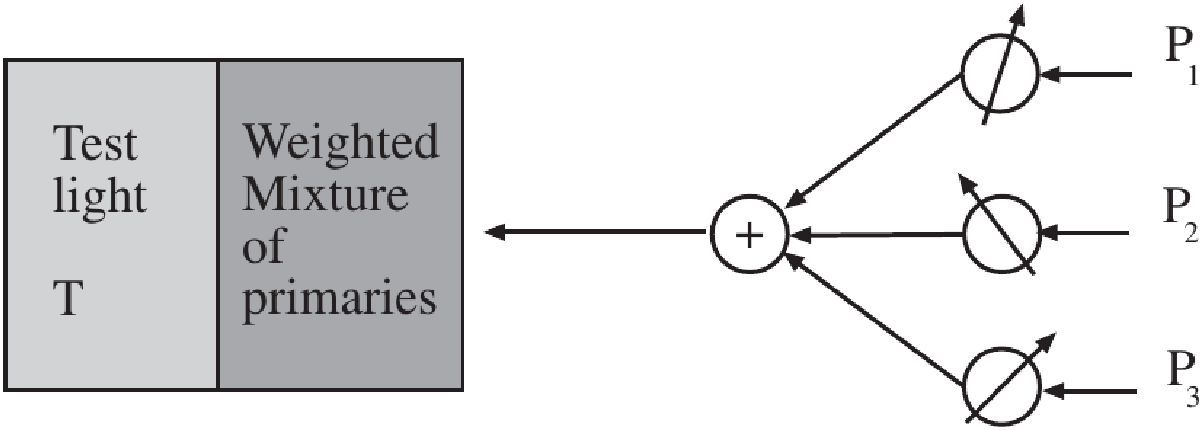
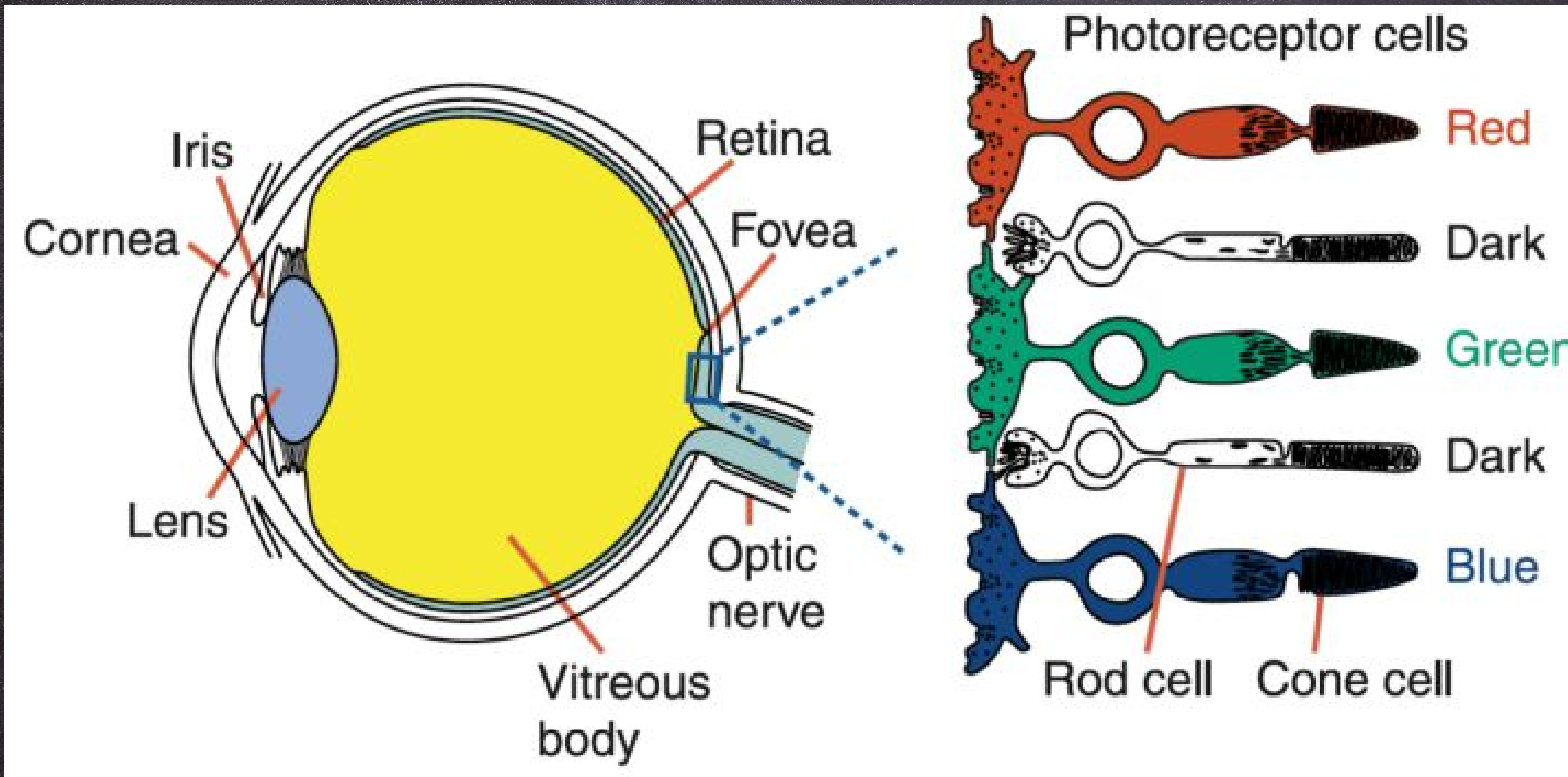


Figure 4.5. Human perception of colour can be studied by asking observers to mix coloured lights to match a test light, shown in a split field. The drawing shows the outline of such an experiment. The observer sees a test light T, and can adjust the amount of each of three primaries in a mixture that is displayed next to the test light. The observer is asked to adjust the amounts so that the mixture looks the same as the test light. The mixture of primaries can be written as $w_1P_1 + w_2P_2 + w_3P_3$; if the mixture matches the test light, then we write $T = w_1P_1 + w_2P_2 + w_3P_3$. It is a remarkable fact that for most people three primaries are sufficient to achieve a match for many colours, and for all colours if we allow subtractive matching (i.e. some amount of some of the primaries is mixed with the test light to achieve a match). Some people will require fewer primaries. Furthermore, most people will choose the same mixture weights to match a given test light.

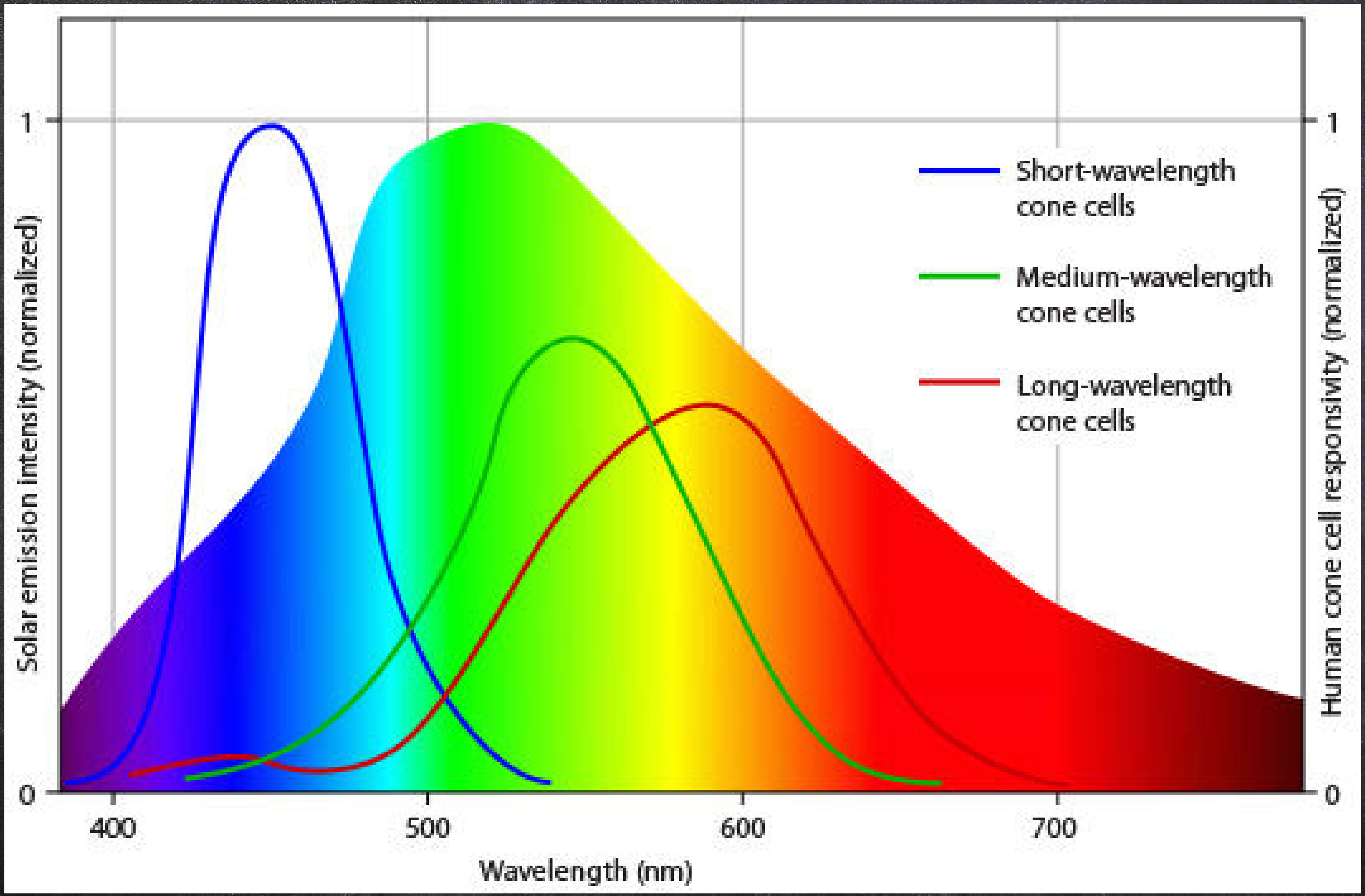
$$
\begin{aligned}
T(\boldsymbol{w}) &= \sum_i w_i P_i \\
T(\boldsymbol{v}) &= \sum_i v_i P_i \\
T(k\boldsymbol{w}) + T(k\boldsymbol{v}) &= kT(\boldsymbol{w} + \boldsymbol{v})
\end{aligned}
$$

$$p_k = \int_\Lambda \sigma_k(\lambda) E(\lambda) \mathrm{d}\lambda$$

彩色视觉


Iris
Cornea
Lens
Retina
Fovea
Optic nerve
Vitreous body
Photoreceptor cells
Red
Dark
Green
Dark
Blue
Rod cell
Cone cell

视锥细胞响应曲线


1
0
400
500
600
700
Wavelength (nm)
Solar emission intensity (normalized)
Human cone cell responsivity (normalized)
1
0
Short-wavelength cone cells
Medium-wavelength cone cells
Long-wavelength cone cells

$$
\begin{aligned}
U(\lambda) &= f_1(\lambda)P_1 + f_2(\lambda)P_2 + f_3(\lambda)P_3 \\
S(\lambda) &= \left(\int_\Lambda f_1(\lambda)S(\lambda)\mathrm{d}\lambda\right)P_1 + \left(\int_\Lambda f_2(\lambda)S(\lambda)\mathrm{d}\lambda\right)P_2 + \left(\int_\Lambda f_3(\lambda)S(\lambda)\mathrm{d}\lambda\right)P_3
\end{aligned}
$$

线性颜色空间

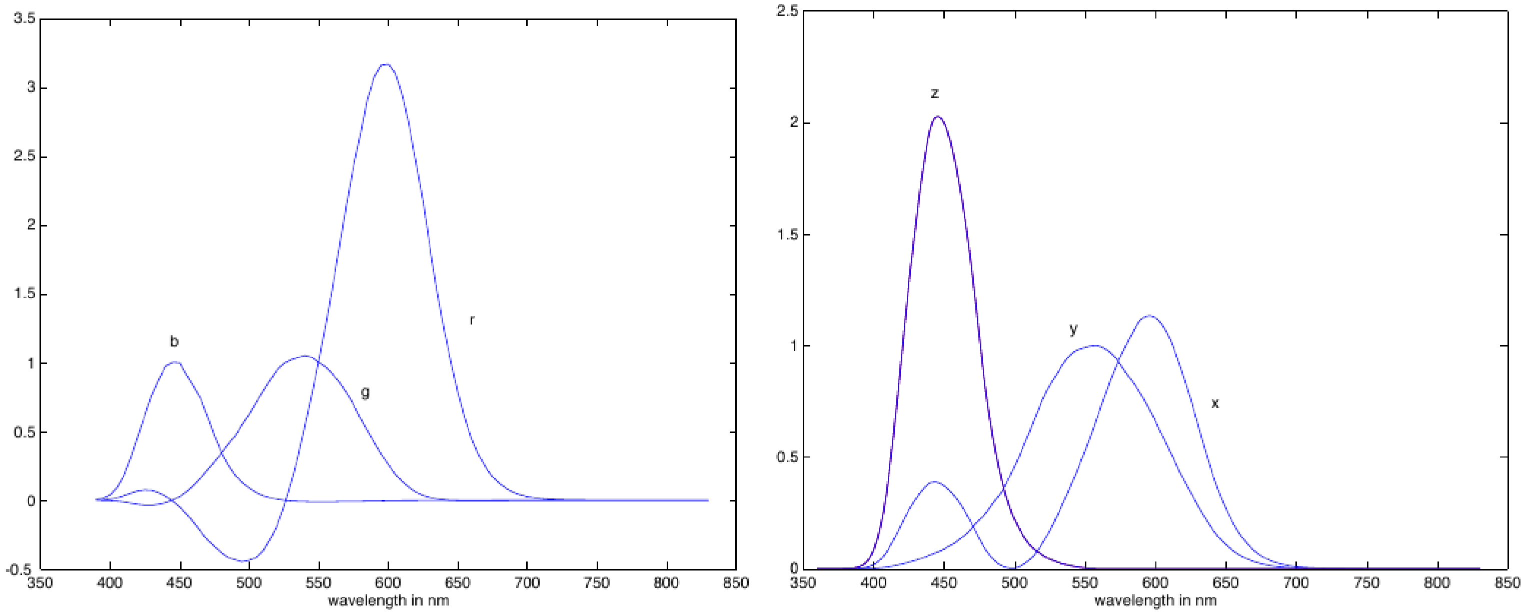


Figure 4.7. On the left, colour matching functions for the primaries for the RGB system. The negative values mean that subtractive matching is required to match lights at that wavelength with the RGB primaries. On the right, colour matching functions for the CIE X, Y and Z primaries; the colourmatching functions are everywhere positive, but the primaries are not real. Figures plotted from data available at `http://www-cvrl.ucsd.edu/index.htm`.

CIE XYZ颜色空间

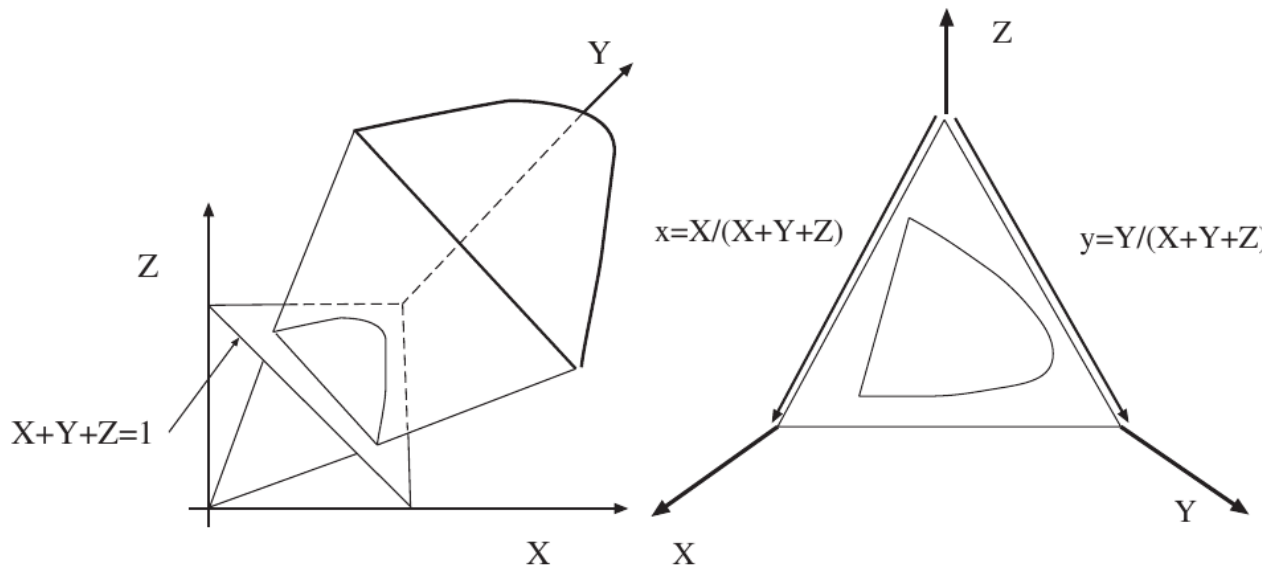


Figure 4.8. The volume of all visible colours in CIE XYZ coordinate space is a cone whose vertex is at the origin. Usually, it is easier to suppress the brightness of a colour — which we can do because to a good approximation perception of colour is linear — and we do this by intersecting the cone with the plane $X + Y + Z = 1$ to get the CIE xy space shown in figures 4.9 and 4.10

CIE xy

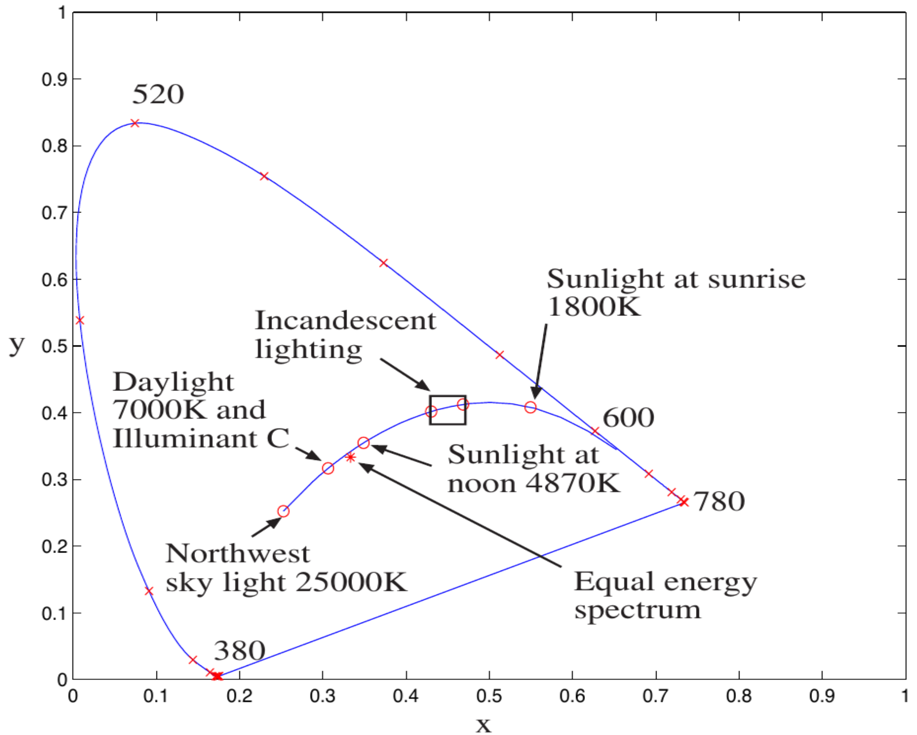


Figure 4.9. The figure shows a constant brightness section of the standard 1931 standard CIE xy colour space. This space has two coordinate axes. The curved boundary of the figure is often known as the spectral locus — it represents the colours experienced when lights of a single wavelength are viewed. The figure shows a locus of colours due to black-body radiators at different temperatures, and a locus of different sky colours. Near the center of the diagram is the neutral point, the colour whose weights are equal for all three primaries. CIE selected the primaries so that this light appears achromatic. Generally, colours that lie further away from the neutral point are more saturated — the difference between deep red and pale pink — and hue — the difference between green and red — as one moves around the neutral point.

Saturation & Hue

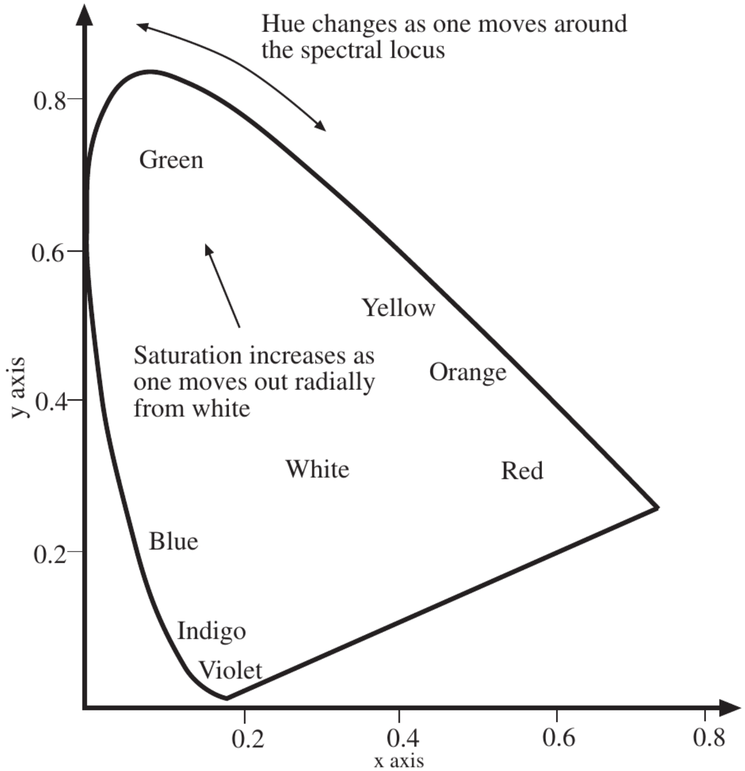


Figure 4.10. The figure shows a constant brightness section of the standard 1931 standard CIE xy colour space, with colour names marked on the diagram. Generally, colours that lie further away from the neutral point are more saturated — the difference between deep red and pale pink — and hue — the difference between green and red — as one moves around the neutral point.

颜色混合

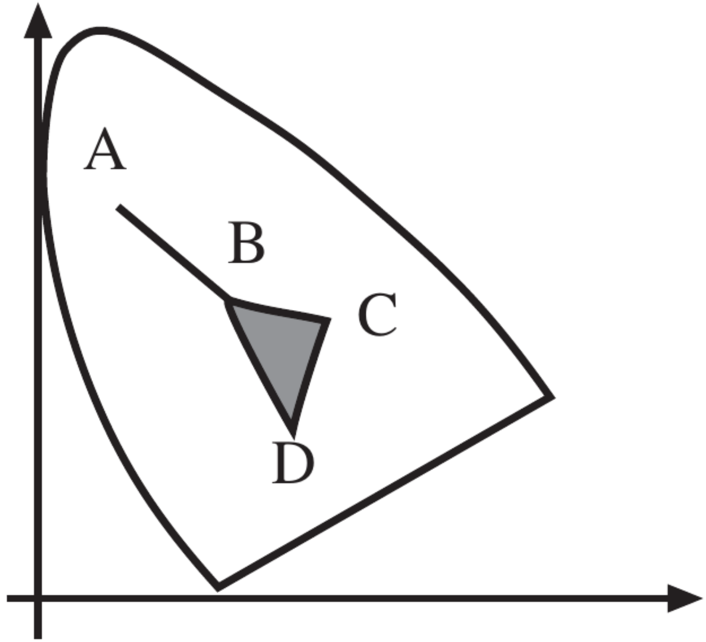
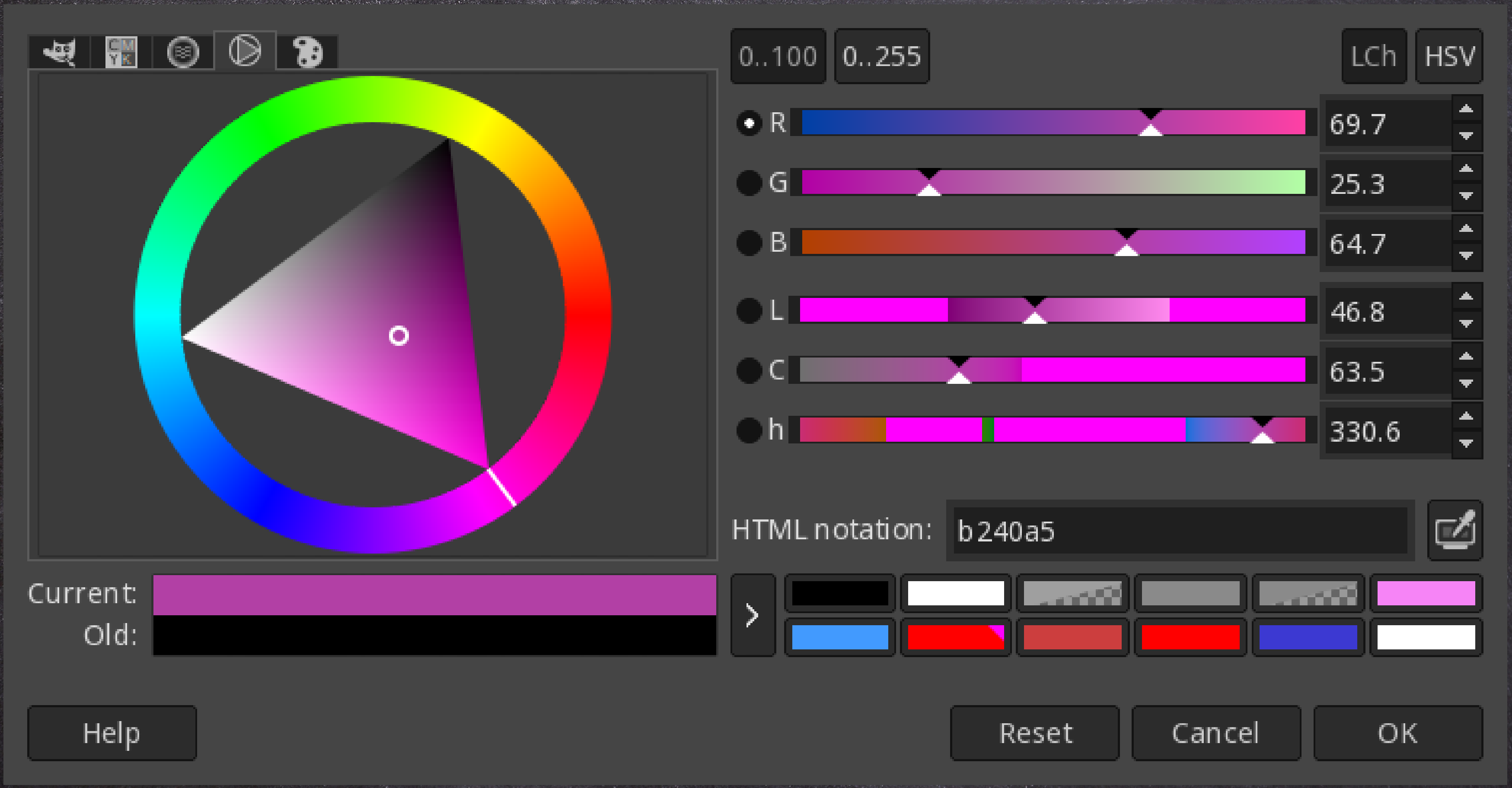


Figure 4.11. The linear model of the colour system allows a variety of useful constructions. If we have two lights whose CIE coordinates are $\boldsymbol{A}$ and $\boldsymbol{B}$ all the colours that can be obtained from non-negative mixtures of these lights are represented by the line segment joining $\boldsymbol{A}$ and $\boldsymbol{B}$. In turn, given $\boldsymbol{B}$, $\boldsymbol{C}$ and $\boldsymbol{D}$, the colours that can by obtained by mixing them lie in the triangle formed by the three points. This is important in the design of monitors — each monitor has only three phosphors, and the more saturated the colour of each phosphor the bigger the set of colours that can be displayed. This also explains why the same colours can look quite different on different monitors. The curvature of the spectral locus gives the reason that no set of three real primaries can display all colours without subtractive matching.

RGB颜色空间


0..100
0..255
LCh
HSV
R 69.7
G 25.3
B 64.7
L 46.8
C 63.5
h 330.6
HTML notation: b240a5
Current:
Old:
Help
Reset
Cancel
OK

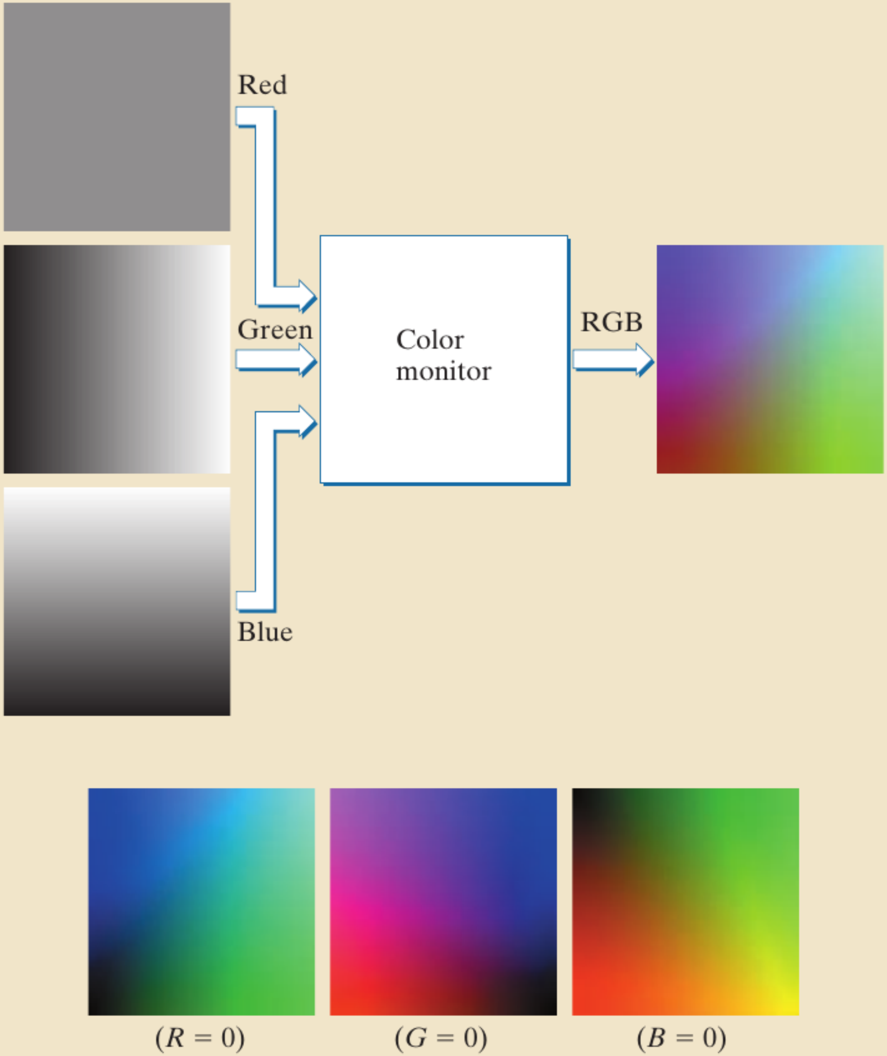
RGB cube

FIGURE 6.8
A 24-bit RGB color cube.

RGB截面

a
b

FIGURE 6.9
(a) Generating the RGB image of the cross-sectional color plane (127, G, B).
(b) The three hidden surface planes in the color cube of Fig. 6.8.



CMYK颜色空间

- cyan (a blue-green colour) = W – R (white – red);
- magenta (a purplish colour) = W – G (white – green)
- and yellow = W – B (white – blue)

Hue, Saturation and Value

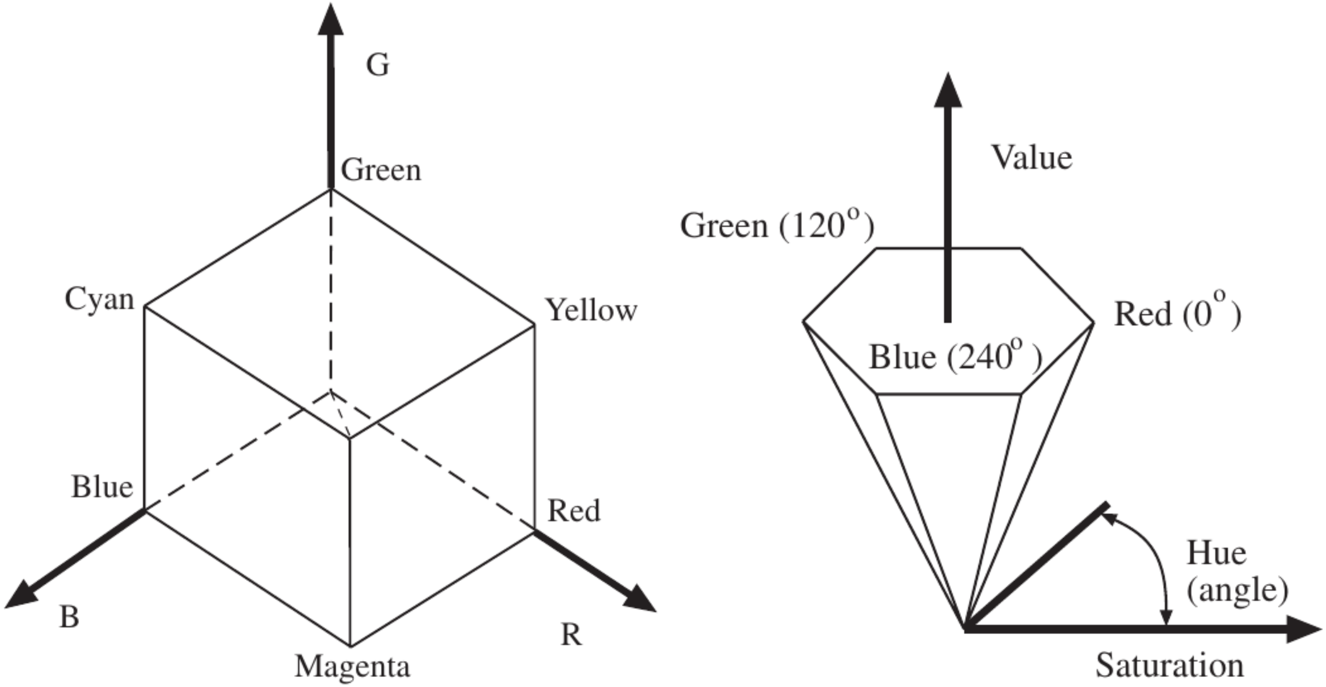
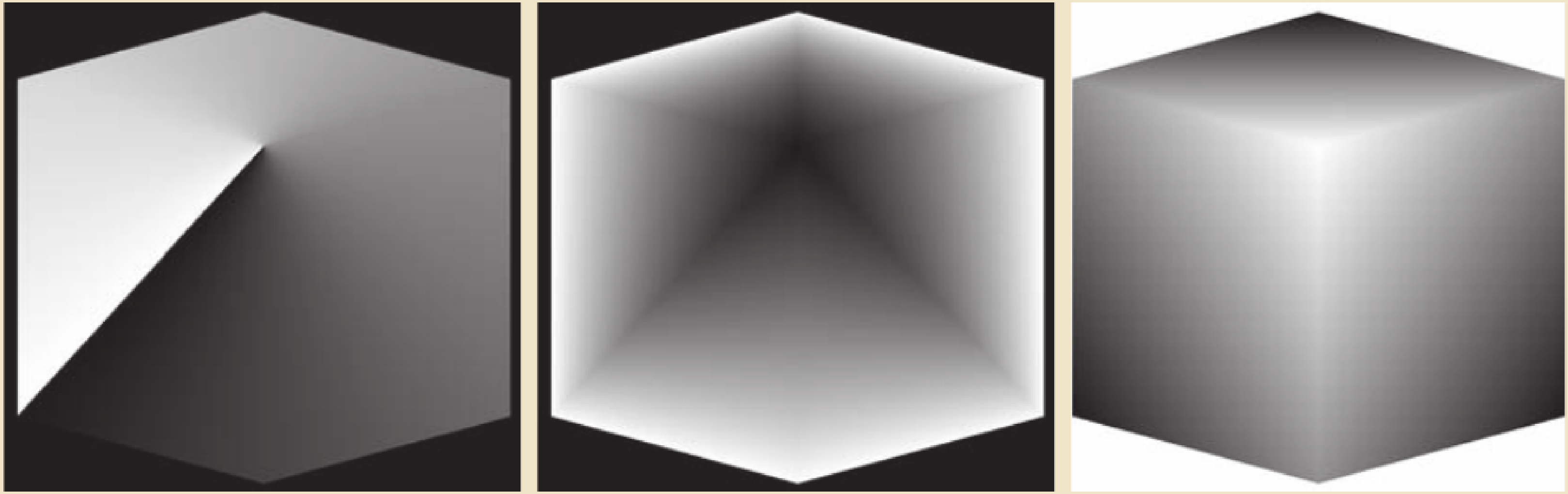


Figure 4.12. On the left, we see the RGB cube; this is the space of all colours that can be obtained by combining three primaries (R, G, and B — usually defined by the colour response of a monitor) with weights between zero and one. It is common to view this cube along its neutral axis — the axis from the origin to the point (1, 1, 1) — to see a hexagon, shown in the middle. This hexagon codes hue (the property that changes as a colour is changed from green to red) as an angle, which is intuitively satisfying. On the right, we see a cone obtained from this cross-section, where the distance along a generator of the cone gives the value (or brightness) of the colour, angle around the cone gives the hue and distance out gives the saturation of the colour.

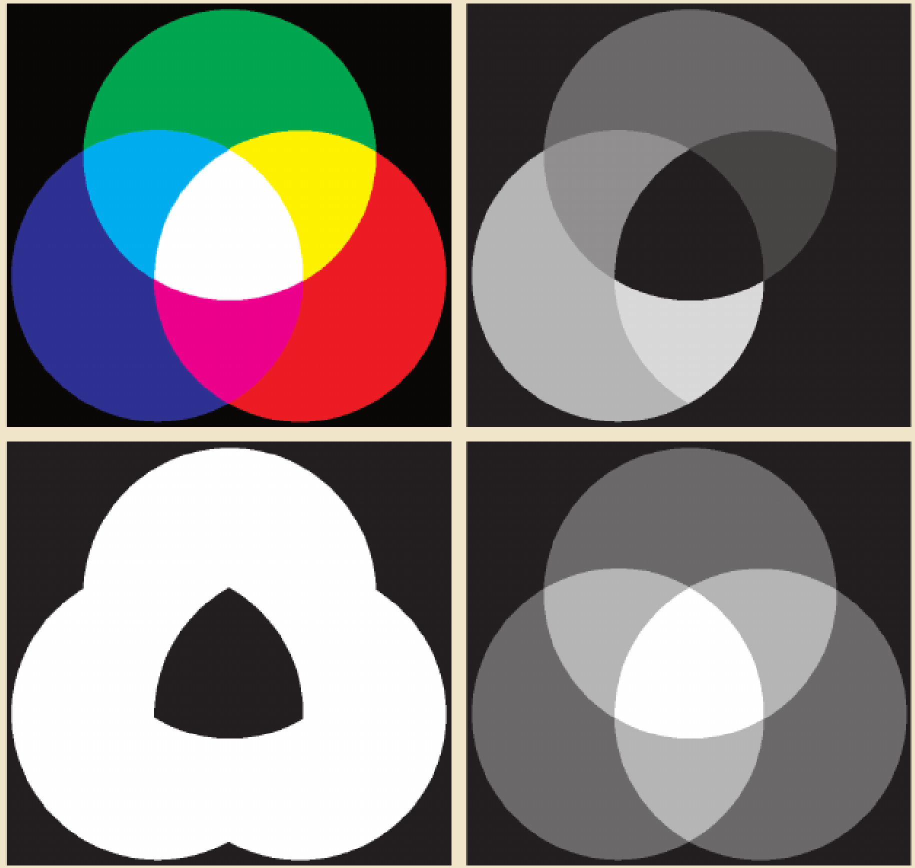
HSI分解



a b c

FIGURE 6.13 HSI components of the image in Fig. 6.8: (a) hue, (b) saturation, and (c) intensity images.

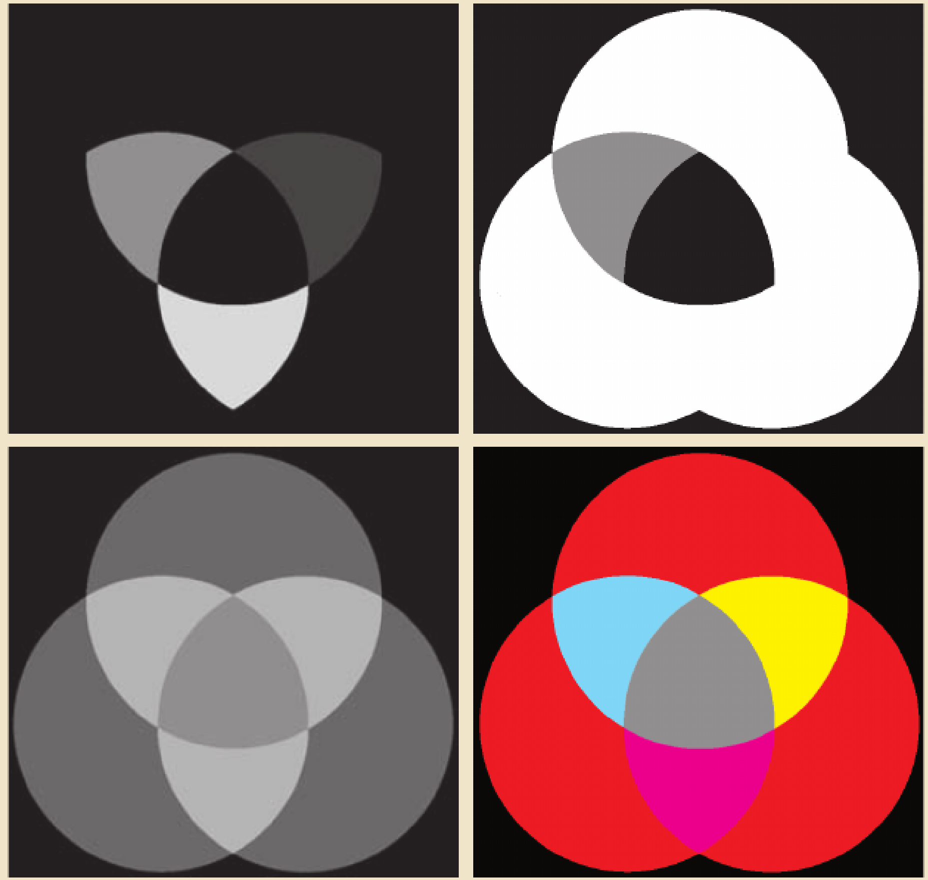
HSI分解



a b
c d

FIGURE 6.14
(a) RGB image and the components of its corresponding HSI image:
(b) hue,
(c) saturation, and
(d) intensity.

HSI合成



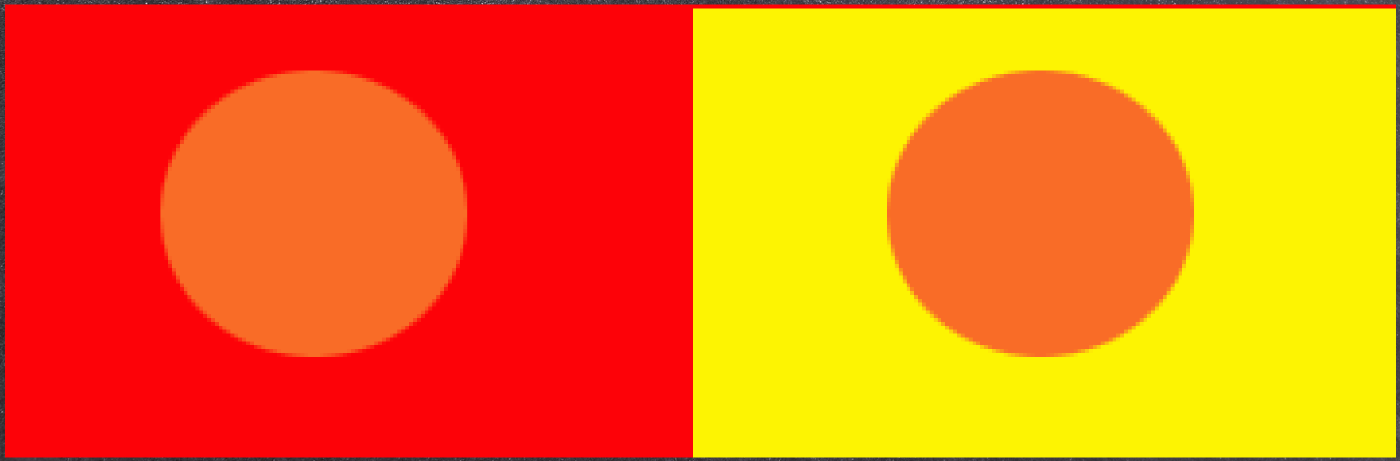
a b
c d

FIGURE 6.15
(a)-(c) Modified HSI component images.
(d) Resulting RGB image. (See Fig. 6.14 for the original HSI images.)

CIE LAB颜色空间

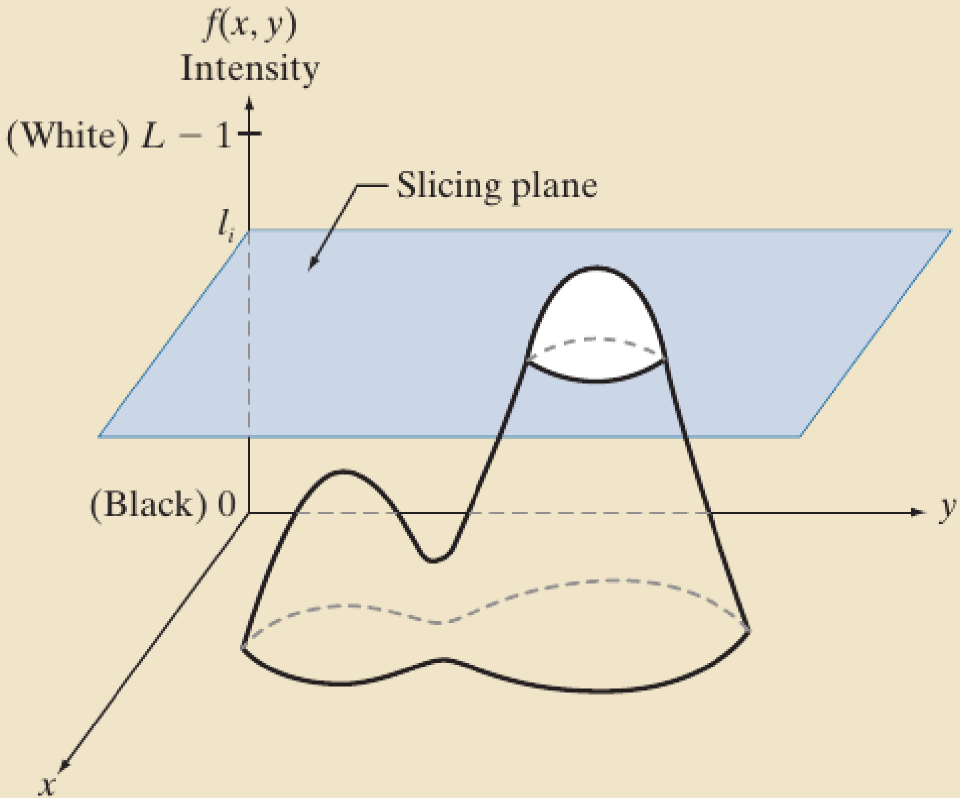
$$
\begin{aligned}
L^* &= 116\left(\frac{Y}{Y_n}\right)^{\frac{1}{3}} - 16 \\
a^* &= 500\left(\left(\frac{X}{X_n}\right)^{\frac{1}{3}} - \left(\frac{Y}{Y_n}\right)^{\frac{1}{3}}\right) \\
b^* &= 200\left(\left(\frac{Y}{Y_n}\right)^{\frac{1}{3}} - \left(\frac{Z}{Z_n}\right)^{\frac{1}{3}}\right)
\end{aligned}
$$

颜色恒常性



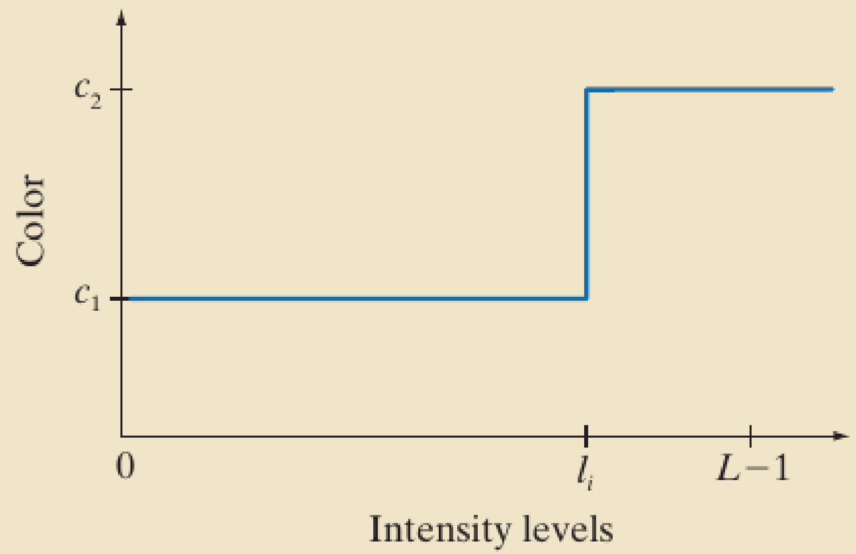
灰度分层

FIGURE 6.16 Graphical interpretation of the intensity-slicing technique.



灰度分层

FIGURE 6.17
An alternative representation of the intensity-slicing technique.



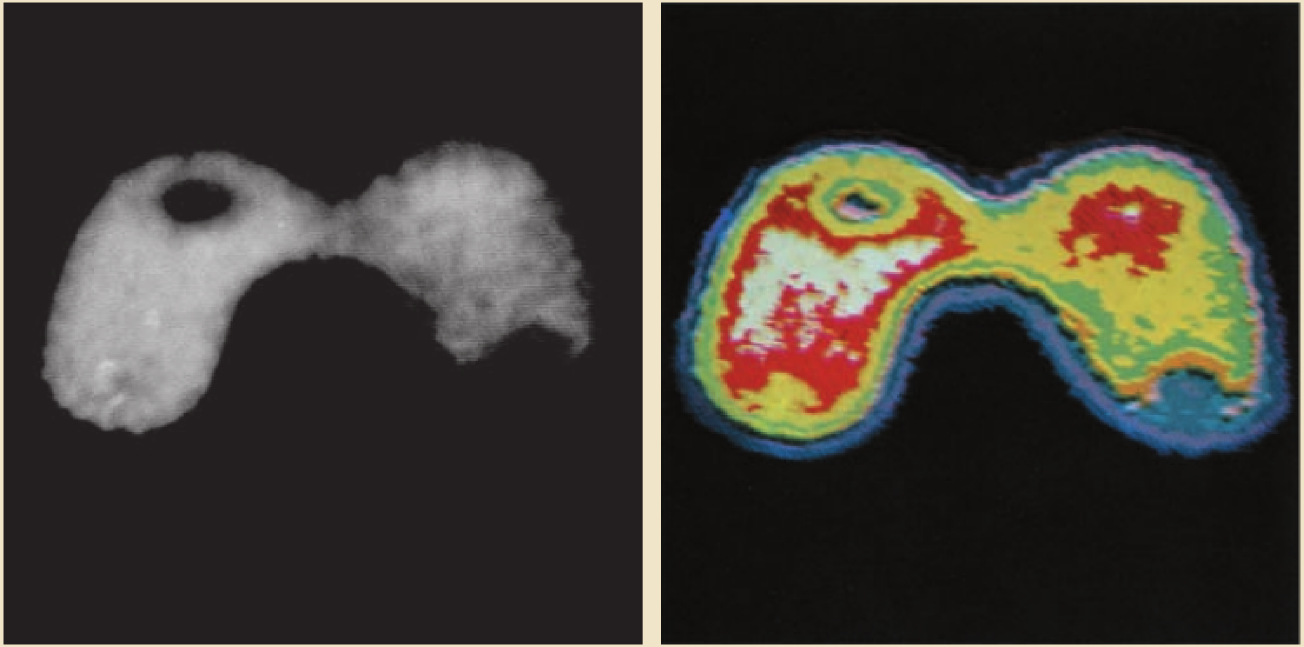
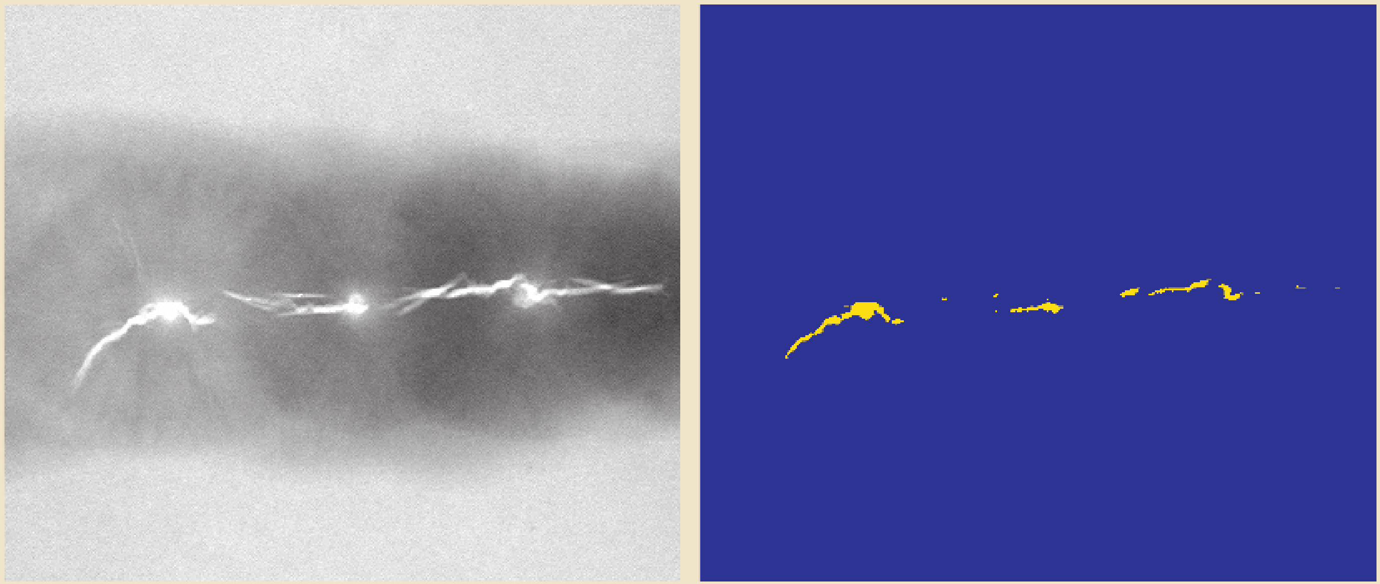


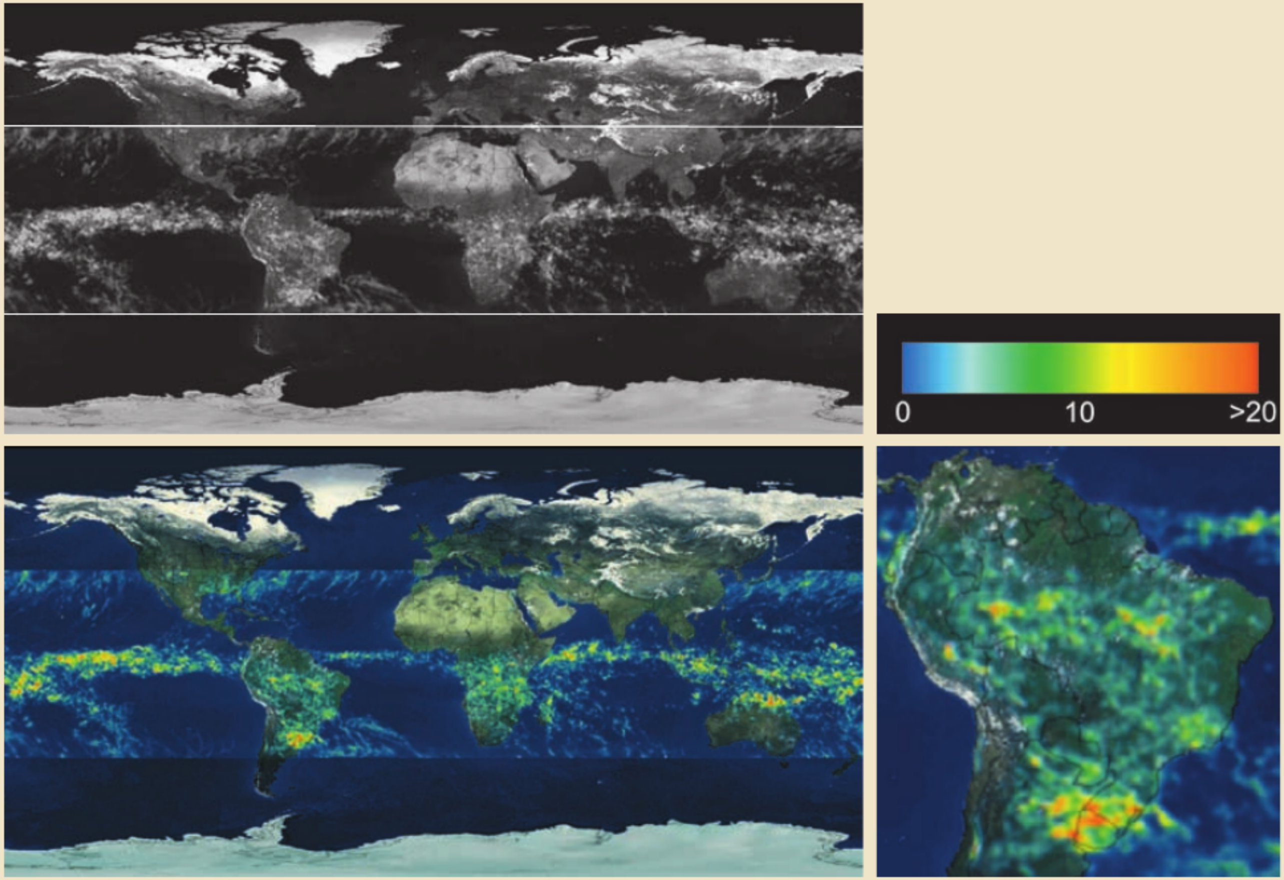
FIGURE 6.18 (a) Grayscale image of the Picker Thyroid Phantom. (b) Result of intensity slicing using eight colors. (Courtesy of Dr. J. L. Blankenship, Oak Ridge National Laboratory.)

灰度分层与伪彩色



a b

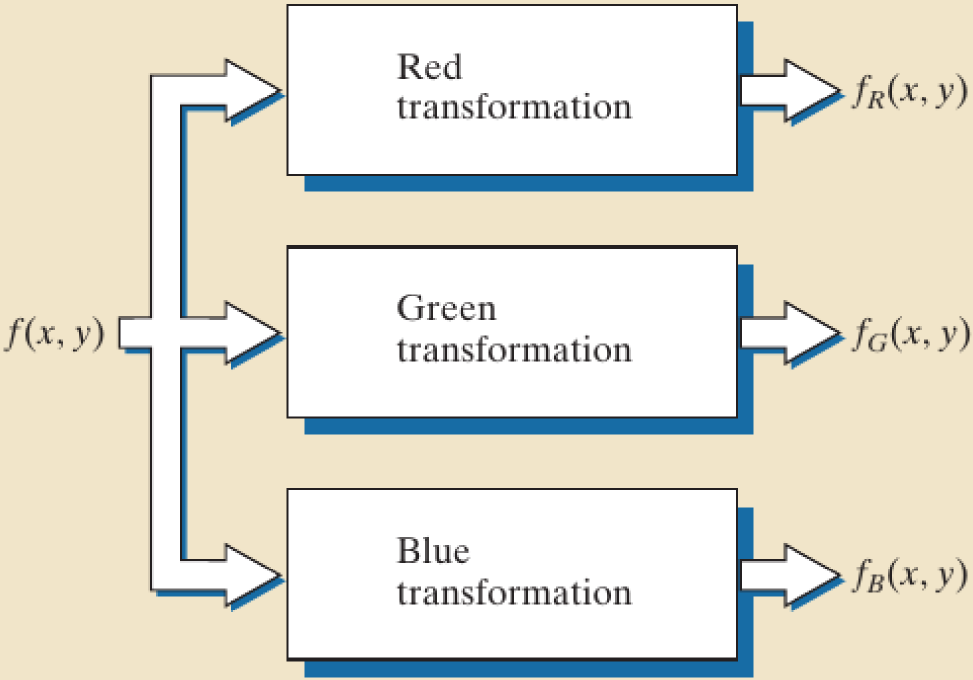
FIGURE 6.19
(a) X-ray image of a weld.
(b) Result of color coding. (Original image courtesy of X-TEK Systems, Ltd.)

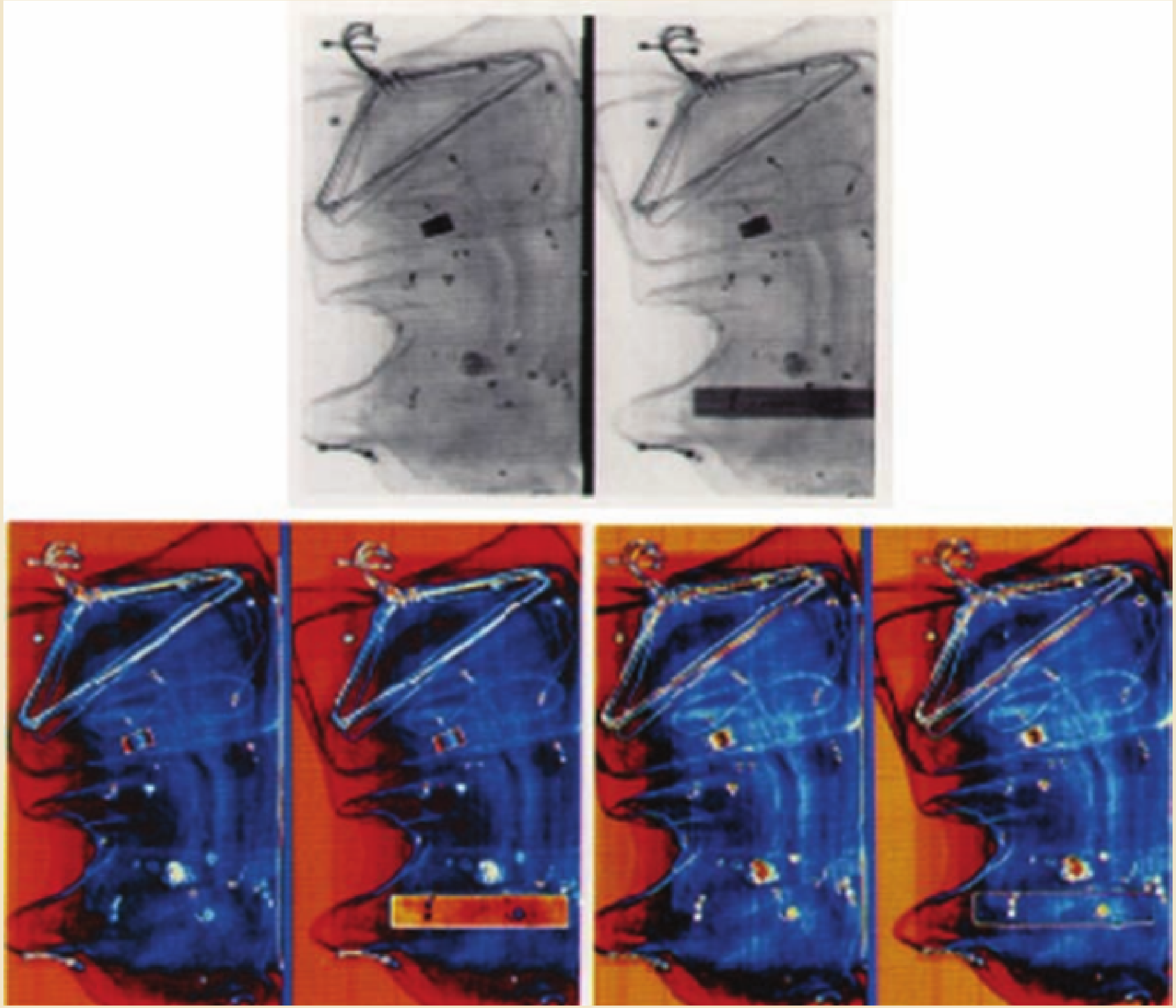


a b
c d

FIGURE 6.20 (a) Grayscale image in which intensity (in the horizontal band shown) corresponds to average monthly rainfall. (b) Colors assigned to intensity values. (c) Color-coded image. (d) Zoom of the South American region. (Courtesy of NASA.)

FIGURE 6.21 Functional block diagram for pseudocolor image processing. Images f_R, f_G, and f_B are fed into the corresponding red, green, and blue inputs of an RGB color monitor.





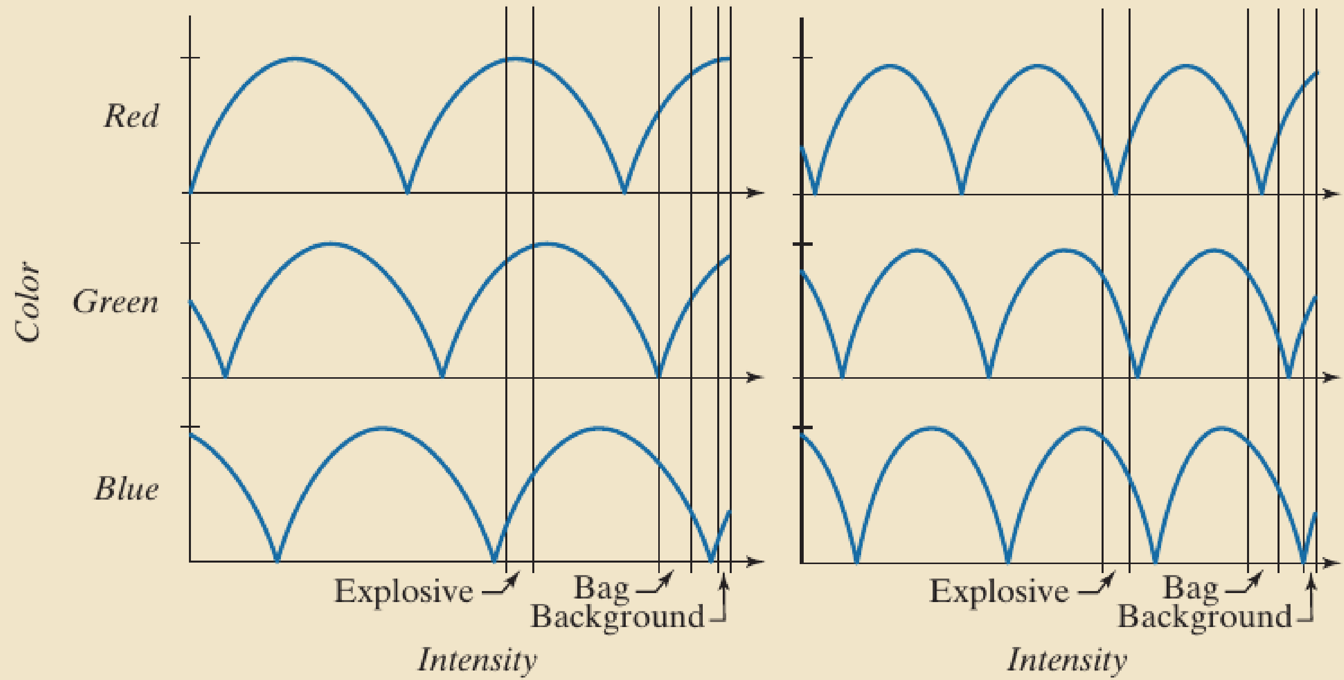
a
b c

FIGURE 6.22 Pseudocolor enhancement by using the gray level to color transformations in Fig. 6.23. (Original image courtesy of Dr. Mike Hurwitz, Westinghouse.)

变换函数

a b

FIGURE 6.23 Transformation functions used to obtain the pseudocolor images in Fig. 6.22.



多光谱

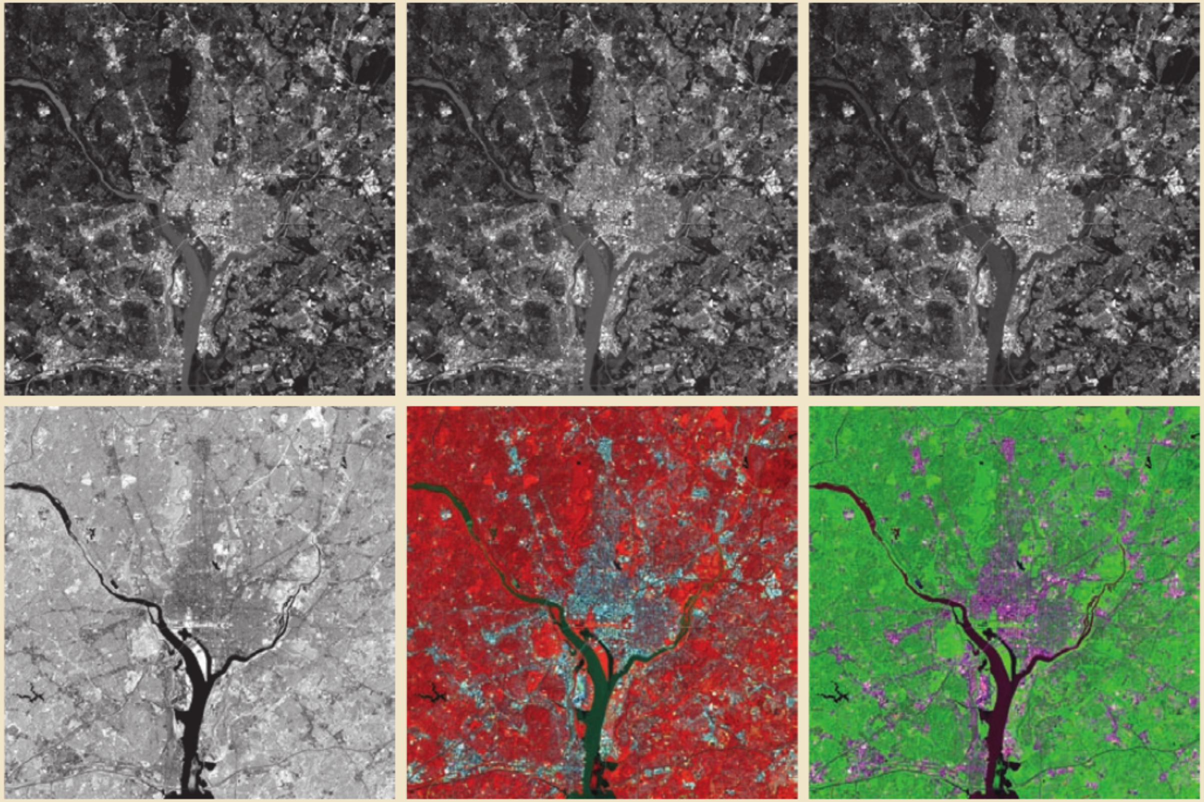
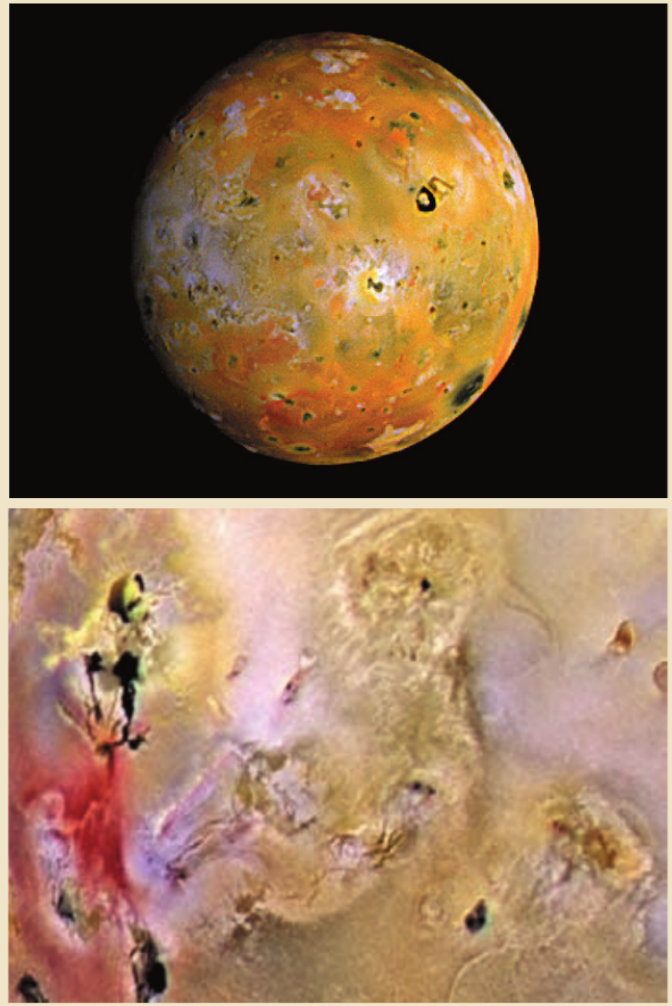


FIGURE 6.25 (a)–(d) Red (R), green (G), blue (B), and near-infrared (IR) components of a LANDSAT multispectral image of the Washington, D.C. area. (e) RGB color composite image obtained using the IR, G, and B component images. (f) RGB color composite image obtained using the R, IR, and B component images. (Original multispectral images courtesy of NASA.)

Jupiter Moon Io

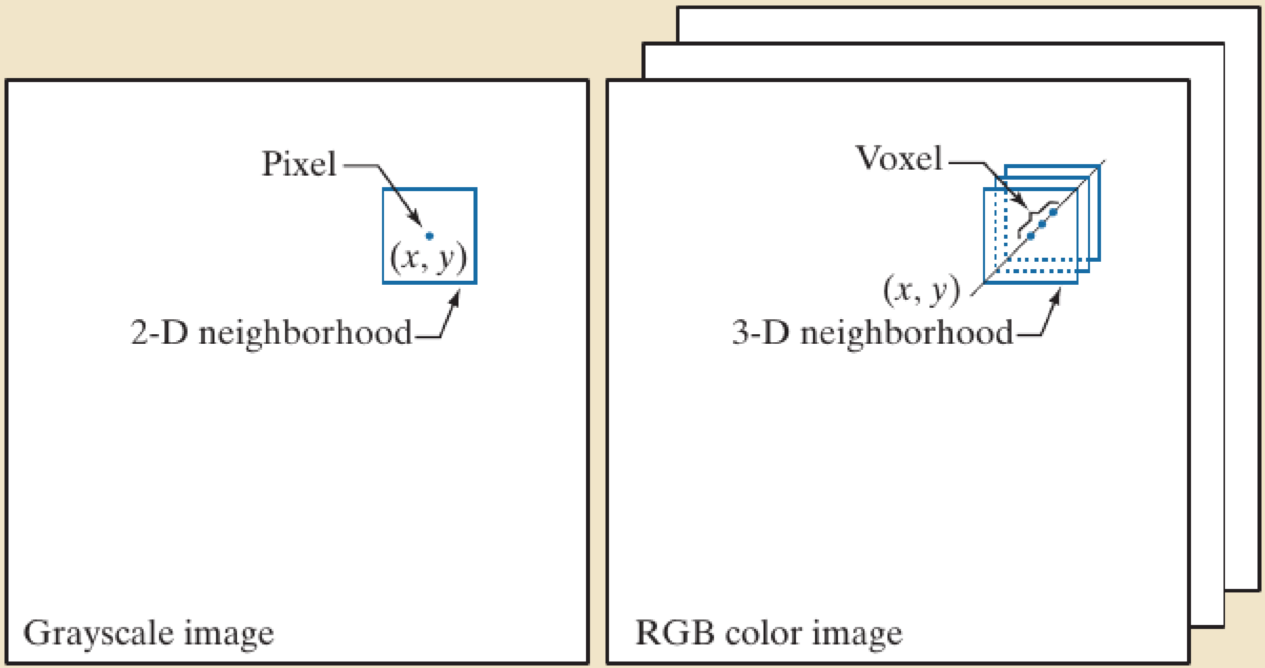
a
b

FIGURE 6.26
(a) Pseudocolor rendition of Jupiter Moon Io. (b) A close-up. (Courtesy of NASA.)



a b

FIGURE 6.27 Spatial neighborhoods for grayscale and RGB color images. Observe in (b) that a *single* pair of spatial coordinates, (x, y), addresses the same spatial location in all three images.



彩色分量

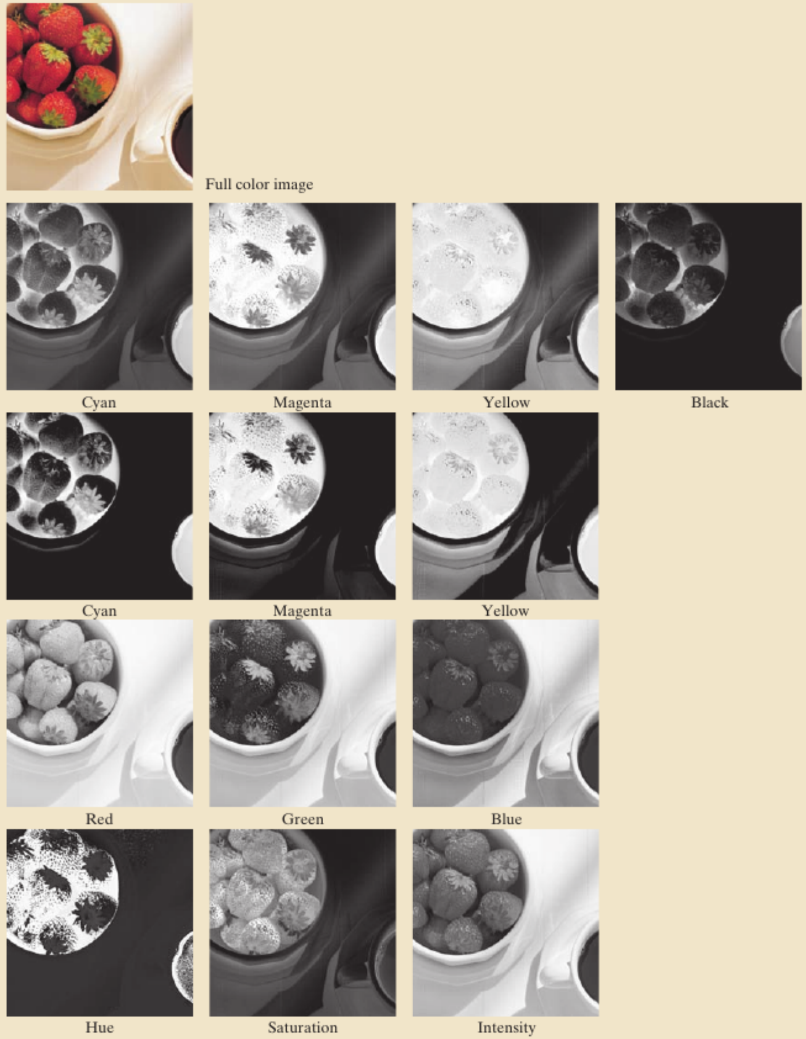


FIGURE 6.28 A full-color image and its various color-space components. (Original image courtesy of MedData Interactive.)

亮度调节

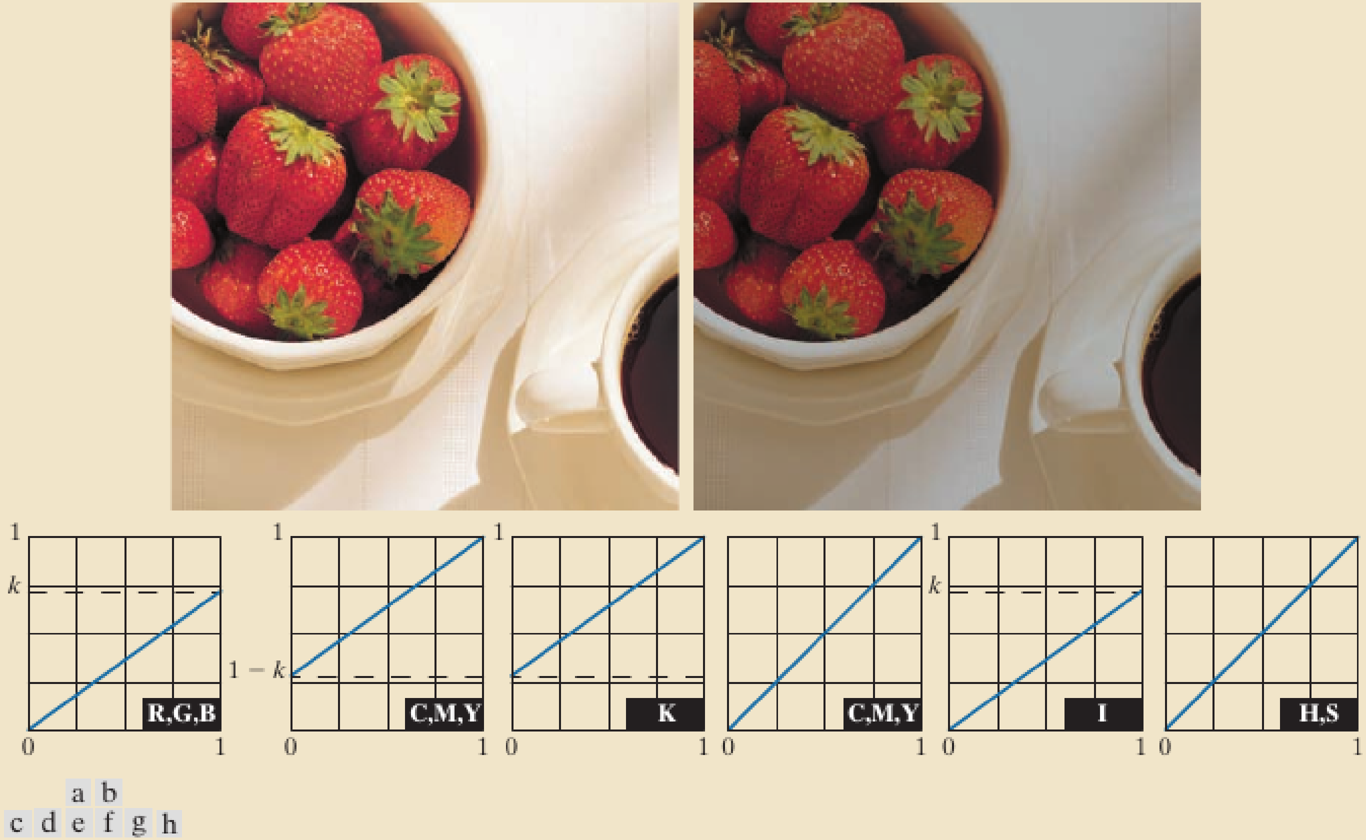
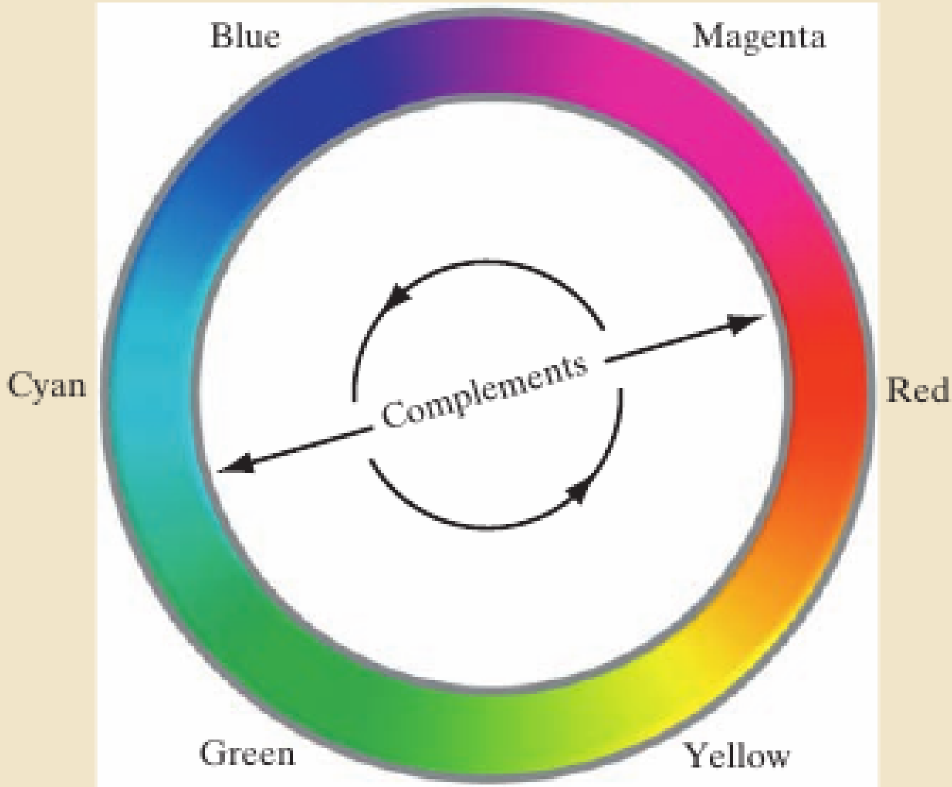


FIGURE 6.29 Adjusting the intensity of an image using color transformations. (a) Original image. (b) Result of decreasing its intensity by 30% (i.e., letting $k = 0.7$). (c) The required RGB mapping function. (d)–(e) The required CMYK mapping functions. (f) The required CMY mapping function. (g)–(h) The required HSI mapping functions. (Original image courtesy of MedData Interactive.)

色环

FIGURE 6.30
Color complements on the color circle.



补色变换

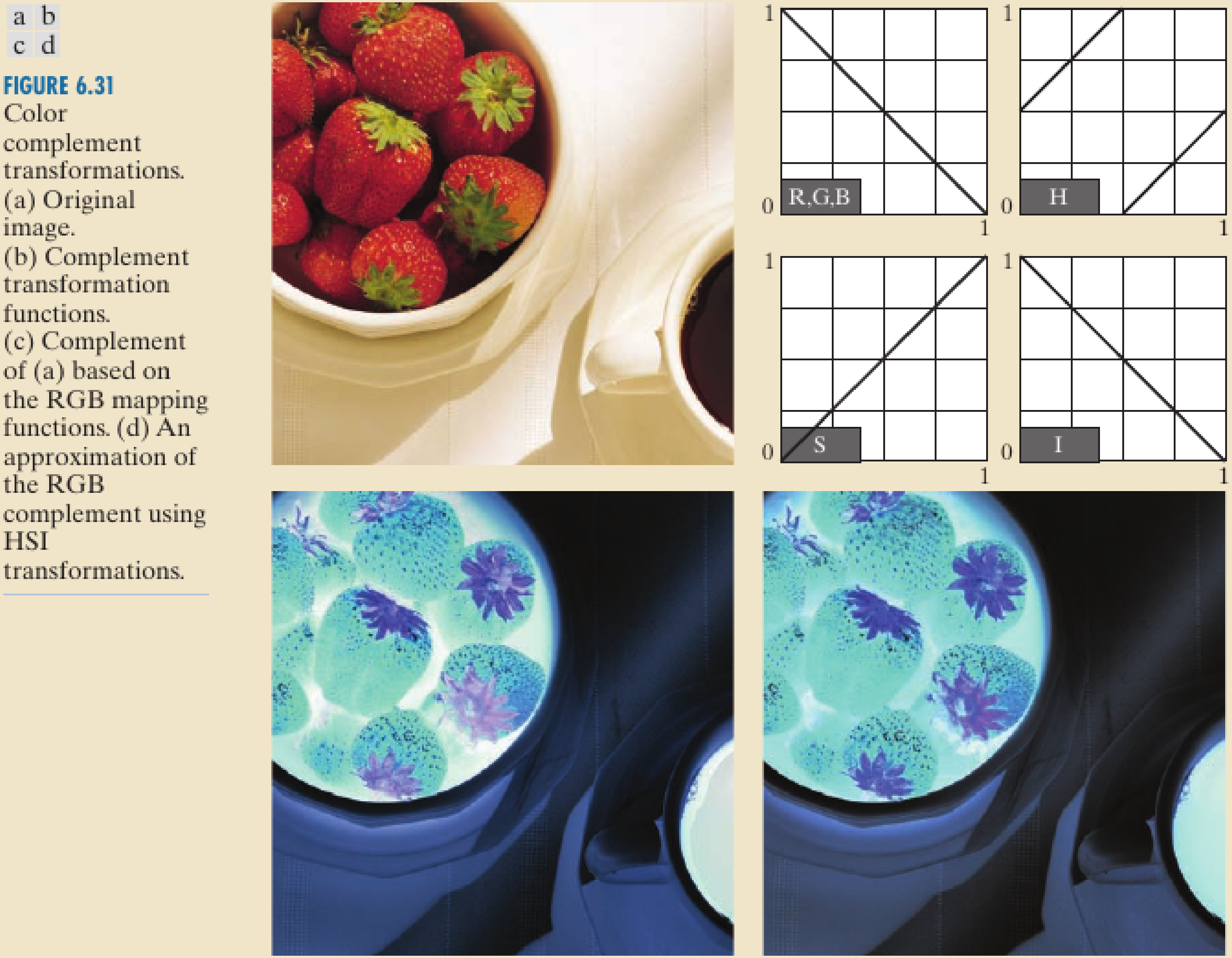
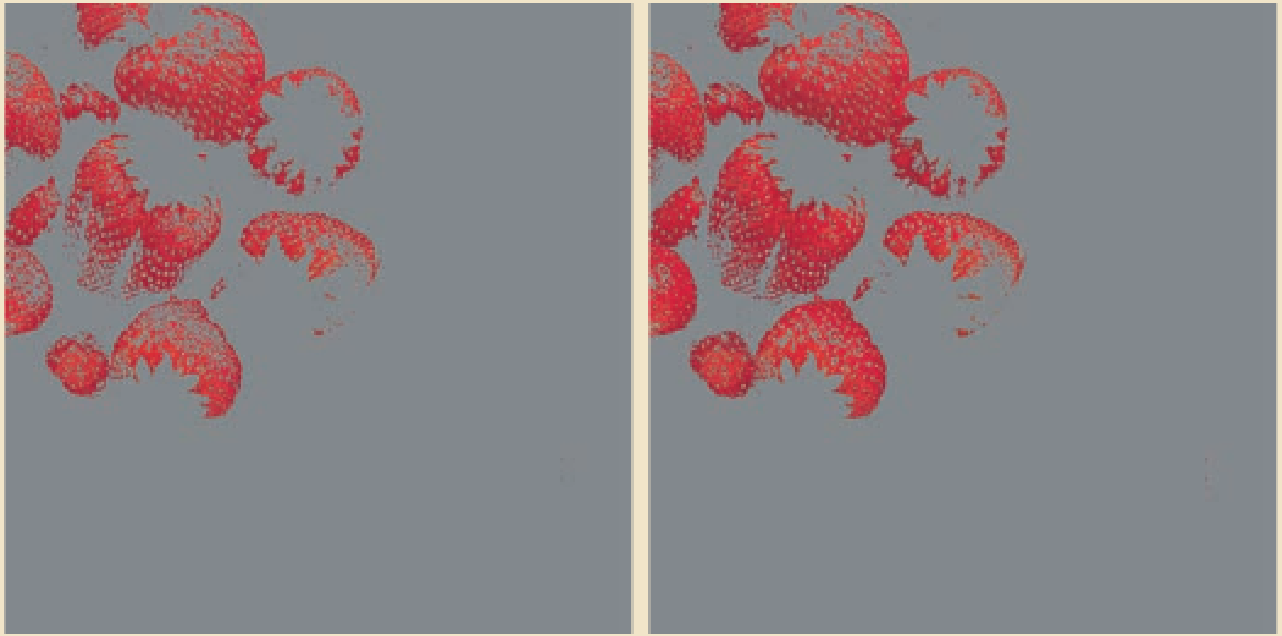


FIGURE 6.31 Color complement transformations. (a) Original image. (b) Complement transformation functions. (c) Complement of (a) based on the RGB mapping functions. (d) An approximation of the RGB complement using HSI transformations.



a b

FIGURE 6.32 Color-slicing transformations that detect (a) reds within an RGB cube of width $W = 0.2549$ centered at $(0.6863, 0.1608, 0.1922)$, and (b) reds within an RGB sphere of radius 0.1765 centered at the same point. Pixels outside the cube and sphere were replaced by color $(0.5, 0.5, 0.5)$.

色调校正

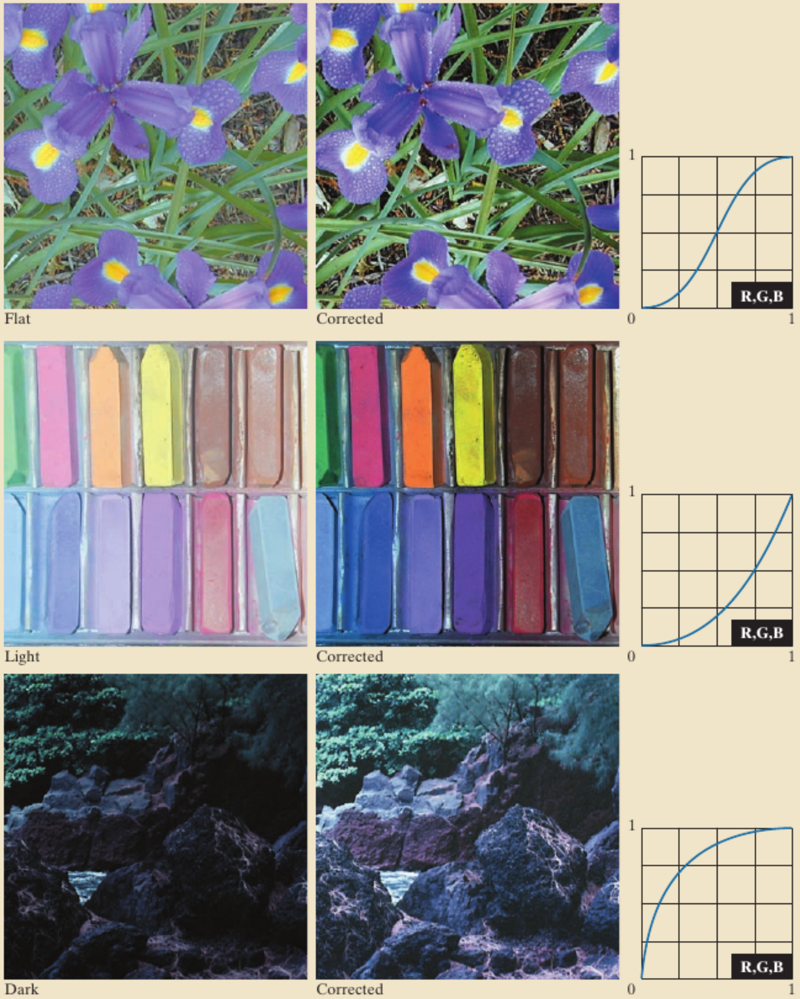
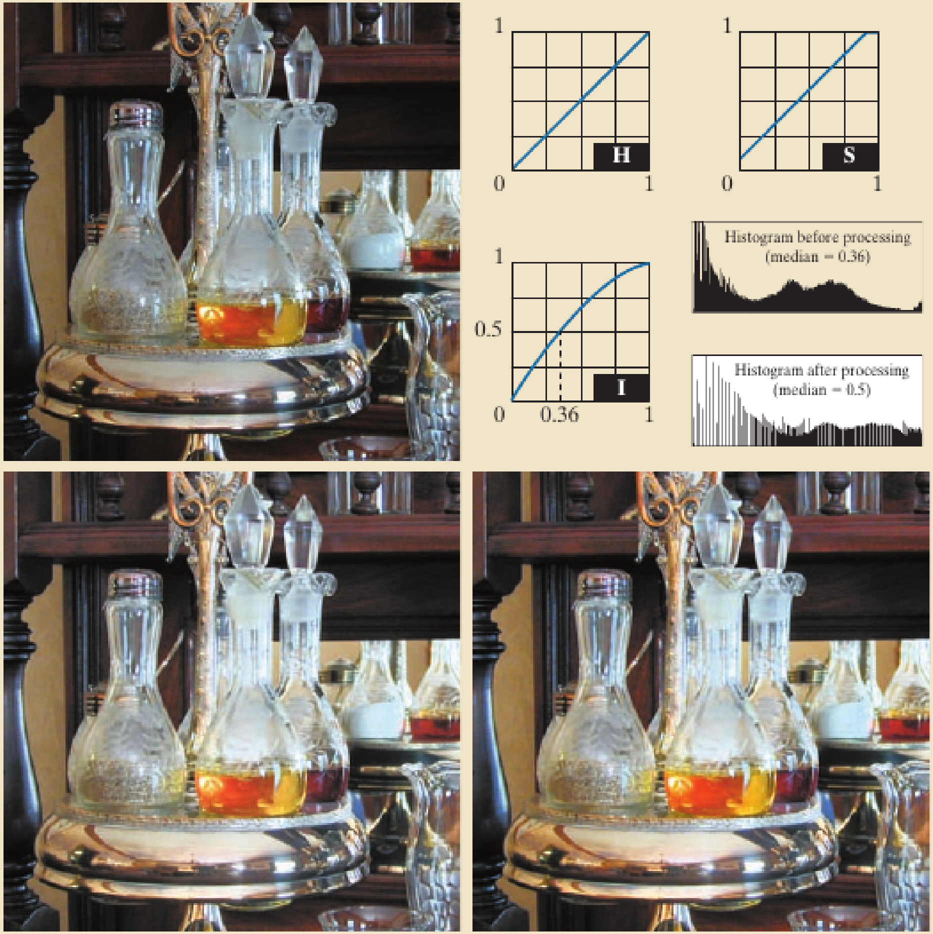


FIGURE 6.33 Tonal corrections for flat, light (high key), and dark (low key) color images. Adjusting the red, green, and blue components equally does not always alter the image hues significantly.

彩色均衡

FIGURE 6.34 Color balancing a CMYK image.

直方图均衡



a b
c d

FIGURE 6.35 Histogram equalization (followed by saturation adjustment) in the HSI color space.

a b
c d

FIGURE 6.36
(a) RGB image.
(b) Red component image.
(c)Green component.
(d) Blue component.

a b c

FIGURE 6.37 HSI components of the RGB color image in Fig. 6.36(a). (a) Hue. (b) Saturation. (c) Intensity.

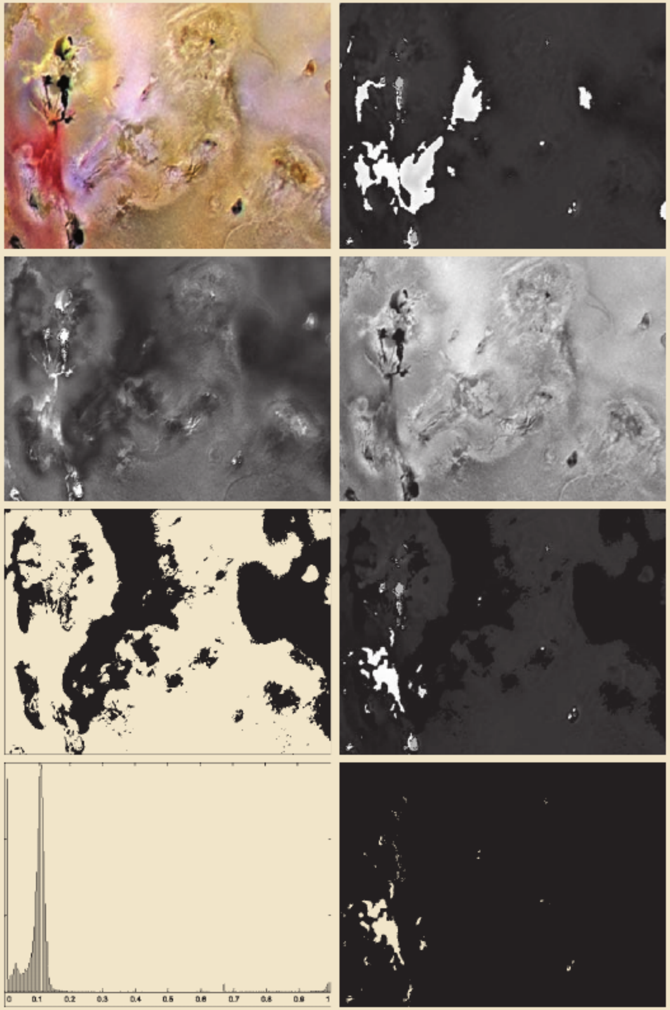
平滑

a b c

FIGURE 6.38 Image smoothing with a 5×5 averaging kernel. (a) Result of processing each RGB component image. (b) Result of processing the intensity component of the HSI image and converting to RGB. (c) Difference between the two results.

a b c

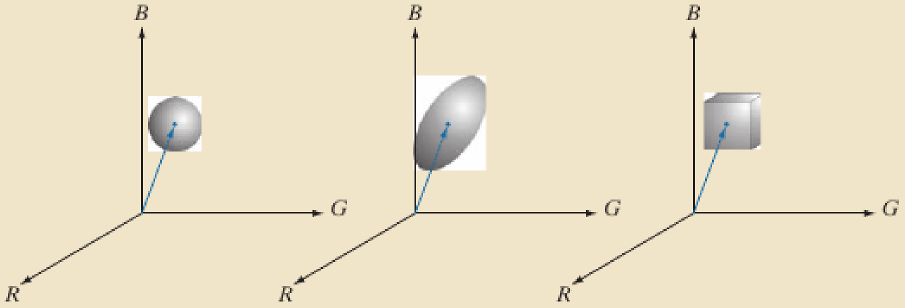
FIGURE 6.39 Image sharpening using the Laplacian. (a) Result of processing each RGB channel. (b) Result of processing the HSI intensity component and converting to RGB. (c) Difference between the two results.



a b
c d
e f
g h

FIGURE 6.40 Image segmentation in HSI space. (a) Original. (b) Hue. (c) Saturation. (d) Intensity. (e) Binary saturation mask (black = 0). (f) Product of (b) and (e). (g) Histogram of (f). (h) Segmentation of red components from (a).

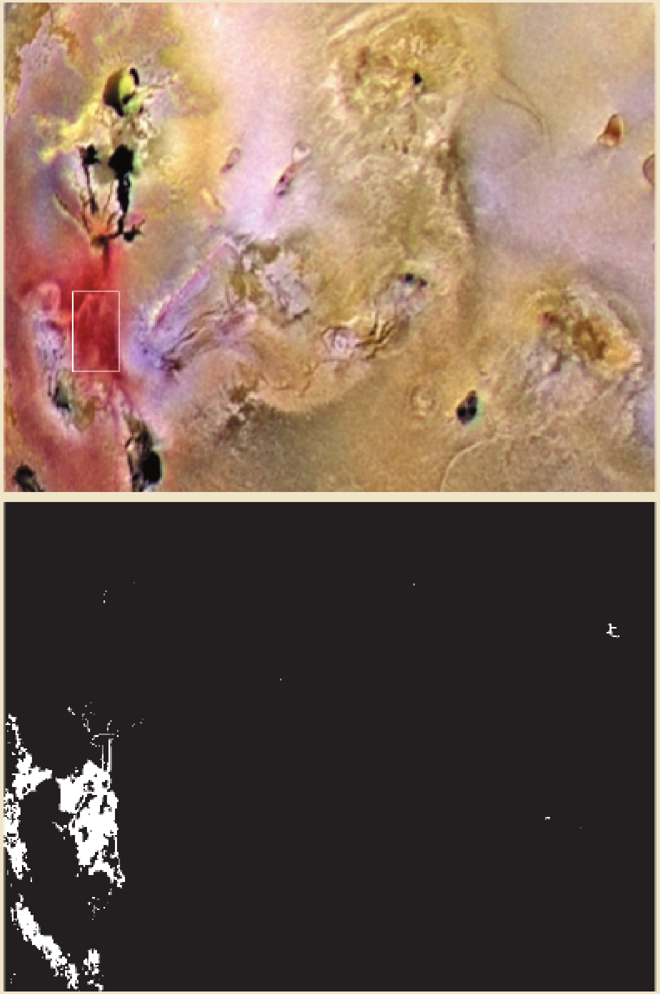
RGB分割



a b c

FIGURE 6.41 Three approaches for enclosing data regions for RGB vector segmentation.

图像分割(RGB)



a
b

FIGURE 6.42 Segmentation in RGB space. (a) Original image with colors of interest shown enclosed by a rectangle. (b) Result of segmentation in RGB vector space. Compare with Fig. 6.40(h).

a b
c d

FIGURE 6.44 (a) RGB image. (b) Gradient computed in RGB color vector space. (c) Gradient image formed by the elementwise sum of three individual gradient images, each computed using the Sobel operators. (d) Difference between (b) and (c).

a b c

FIGURE 6.45 Component gradient images of the color image in Fig. 6.44. (a) Red component, (b) green component, and (c) blue component. These three images were added and scaled to produce the image in Fig. 6.44(c).

a b
c d

FIGURE 6.46
(a)–(c) Red, green, and blue 8-bit component images corrupted by additive Gaussian noise of mean 0 and standard deviation of 28 intensity levels. (d) Resulting RGB image. [Compare (d) with Fig. 6.44(a).]

噪声(HSI)

FIGURE 6.47 HSI components of the noisy color image in Fig. 6.46(d). (a) Hue. (b) Saturation. (c) Intensity.

噪声(单通道)

FIGURE 6.48 (a) RGB image with green plane corrupted by salt-and-pepper noise. (b) Hue component of HSI image. (c) Saturation component. (d) Intensity component.